

Nauta® Mixing Deep Drive

by Chris Paulsworth

- The presentation will start at 2:00 PM Eastern Time (USA)
- Hosokawa panelist will not be able to hear you, however you can submit questions using the dialog box to be answered at the conclusion of the presentation. If your question is not answered, a Hosokawa representative will respond offline.
- This presentation will be recorded and available on the Hosokawa Micron Powder Systems' Website within 48 hours. Please share with colleagues and co-workers.
- Although information presented during this webinar is considered accurate and deemed reliable, no guarantee can be made from this content. Please contact Hosokawa for more information.
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HOSOKAWA MICRON POWDER SYSTEMS

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2. How Nauta® Mixers Mix
3. Component Breakdown
4. Using the Mixer as a Dryer
5. Common Applications
6. Questions and Answers



The beginning

UNITED STATES PATENT OFFICE

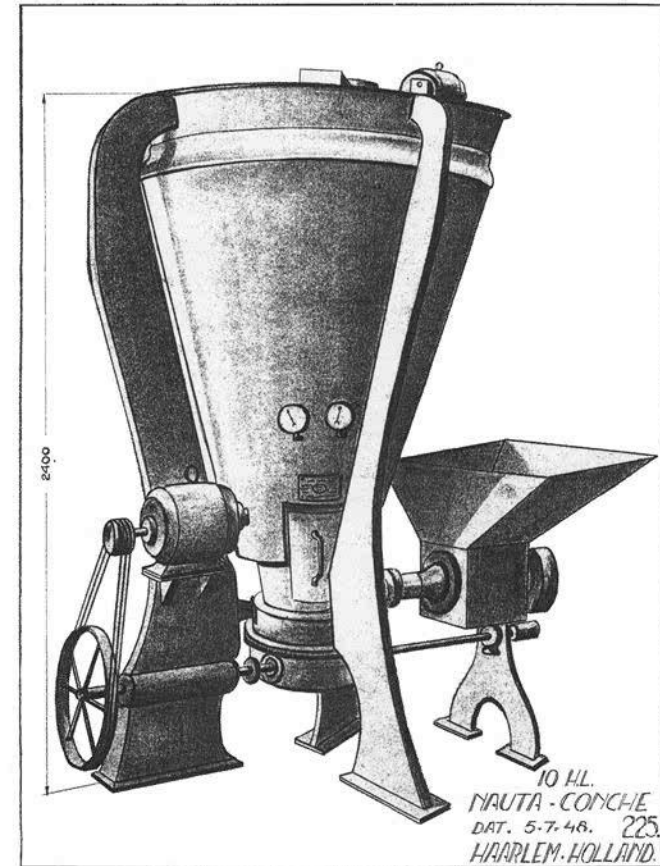
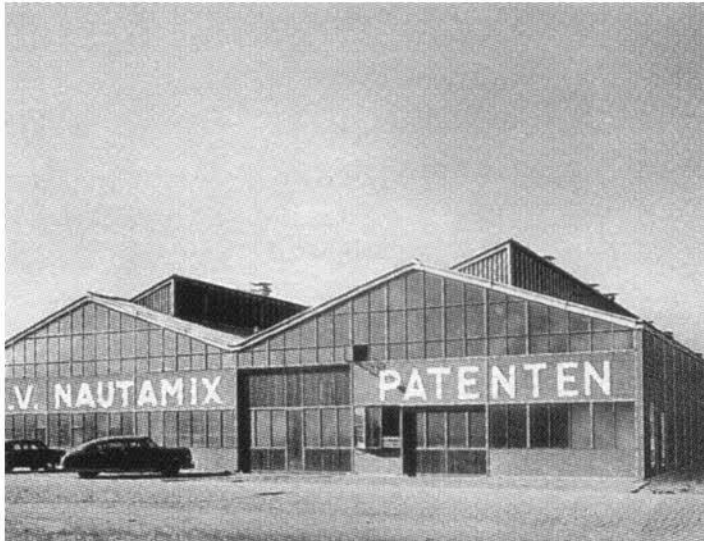
2,345,063

**MIXING APPARATUS, ESPECIALLY ADAPTED
FOR MIXING PULVERIZED MATERIALS**

**Johannes Ewardus Nauta, Overveen, Nether-
lands; vested in the Allen Property Custodian**

**Application October 31, 1940, Serial No. 363,778
In the Netherlands November 13, 1939**

6 Claims. (Cl. 259—40)



HOSOKAWA MICRON POWDER SYSTEMS

The original design

March 28, 1944.

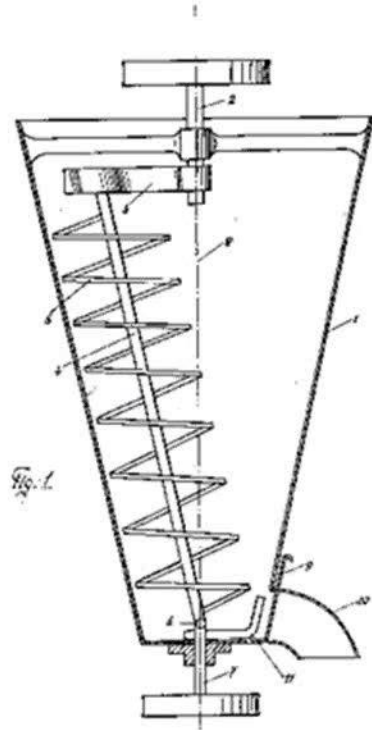
J. E. NAUTA

2,345,063

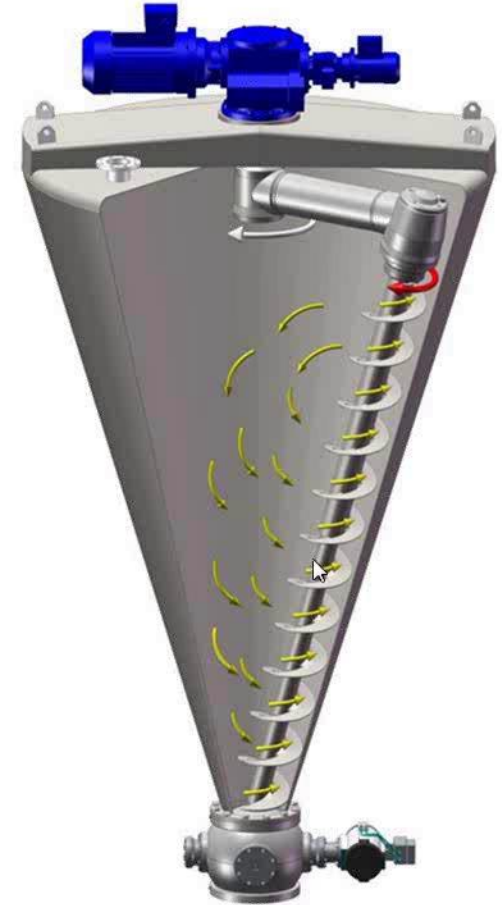
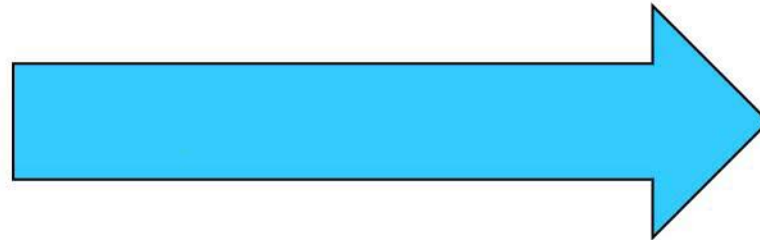
MIXING APPARATUS, ESPECIALLY ADAPTED FOR MIXING FULVERIZED MATERIALS

Filed Oct. 31, 1940

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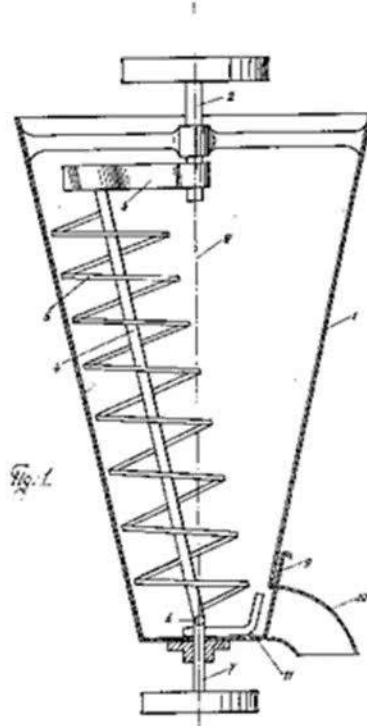
Inventor
Johannes E. Nauta
by *Alfred H. Nauta*
Att'y.



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Two deigns?

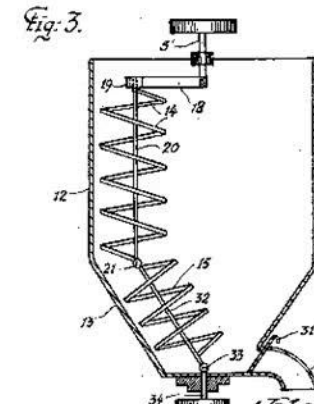
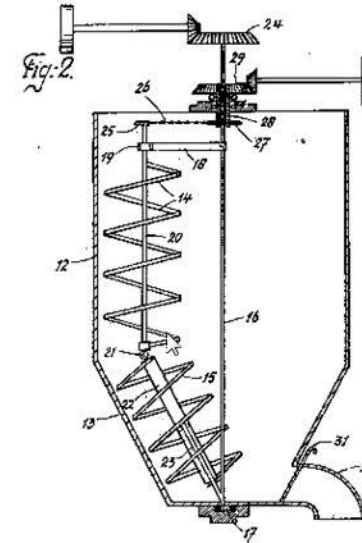
March 28, 1944. J. E. NAUTA 2,345,063
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Filed Oct. 31, 1940 2 Sheets-Sheet 1



Inventor
Johannes E. Nauta
by *[Signature]*
Atty.

&

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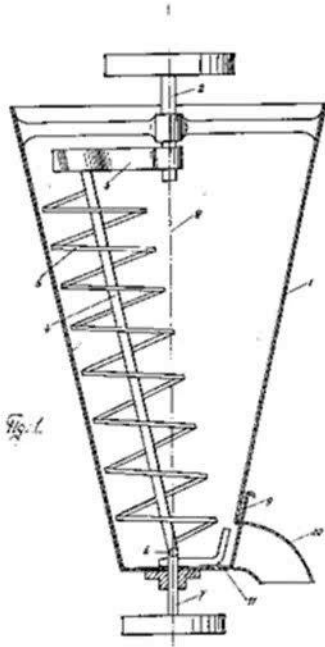
Inventor
Johannes E. Nauta
by *[Signature]*
Atty.



HOSOKAWA MICRON POWDER SYSTEMS

73 years of continuous improvement

March 28, 1944. J. E. NAUTA 2,345,063
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Inventor
Johannes D. Nauta
by *[Signature]*
Att'y.



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Over the years



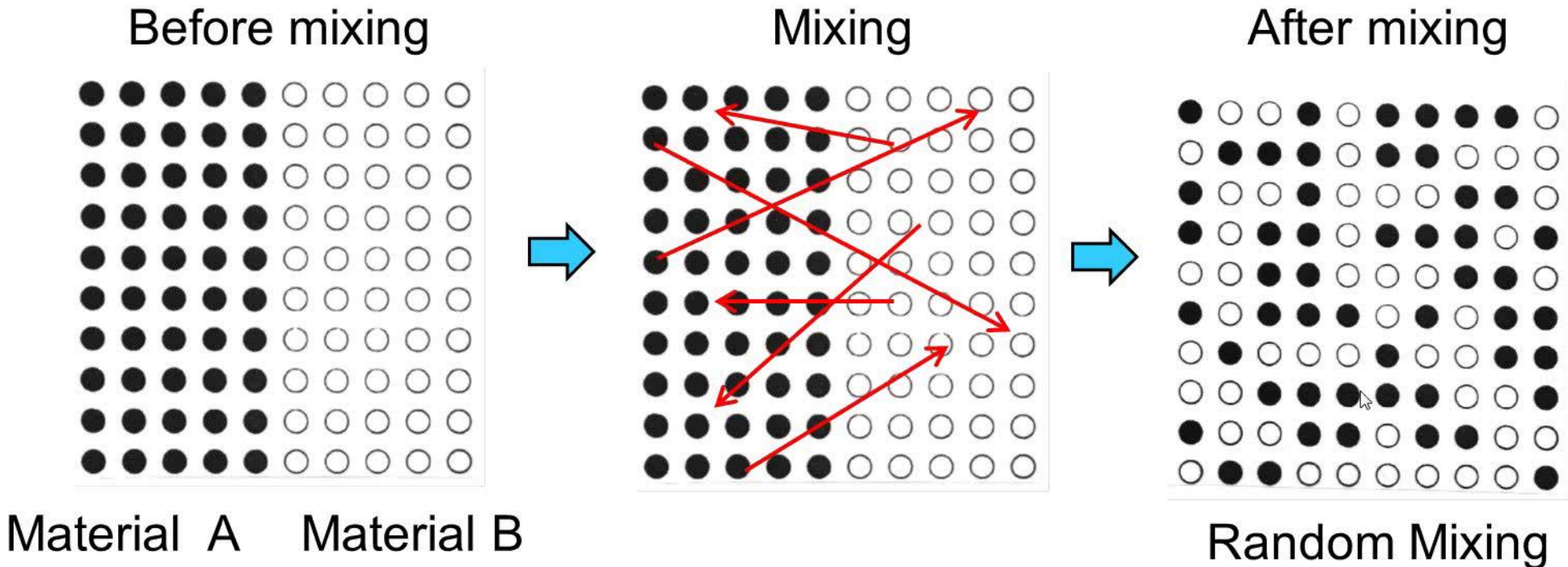
*Nautamix + Machinefabrik Vrieco + ISEM ...
+ Schugi + Hosokawa Micron
= Hosokawa Micron B.V.*



HOSOKAWA MICRON POWDER SYSTEMS

How Mixers Work

The Mixing Process is about transporting particles



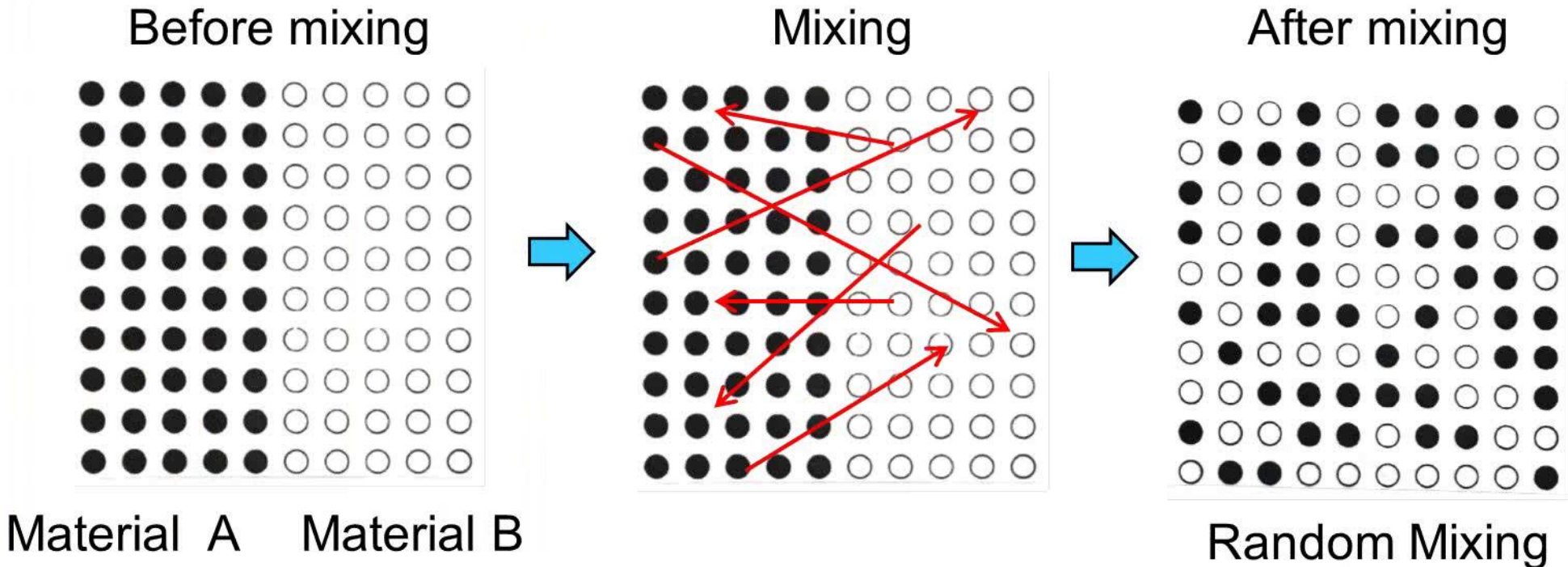
Why Nauta® Mixers?



HOSOKAWA MICRON POWDER SYSTEMS

How Mixers Work

The Mixing Process is about transporting particles

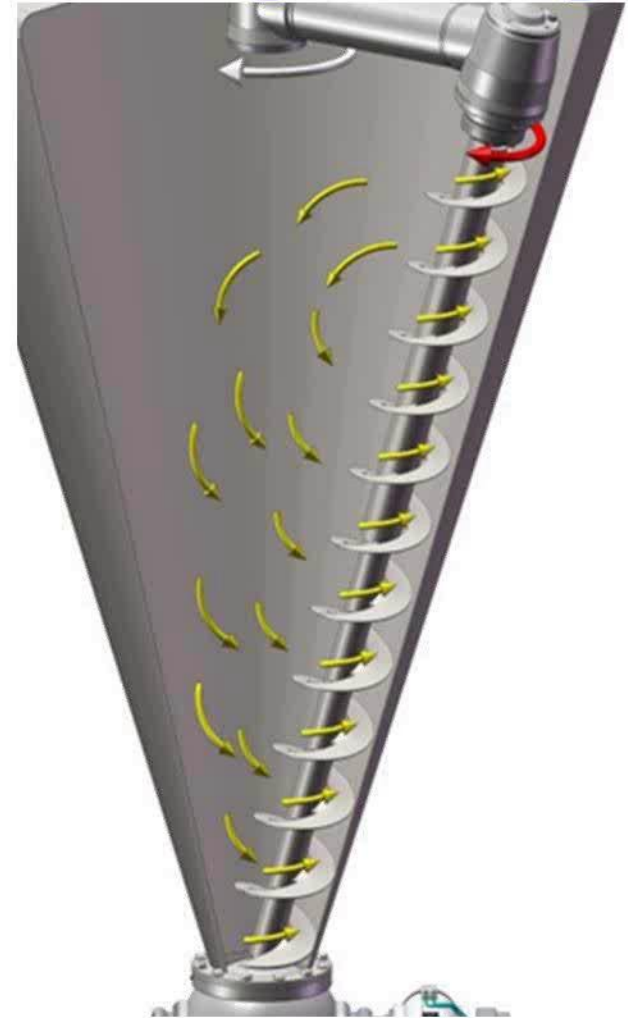
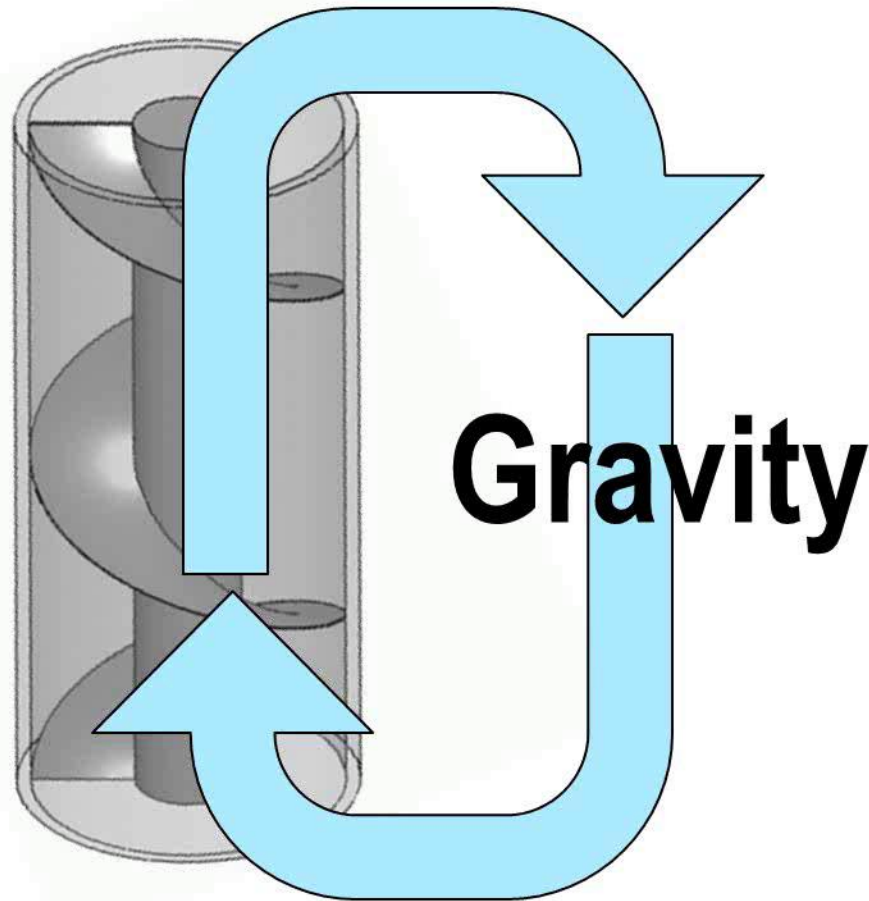


Why Nauta® Mixers?

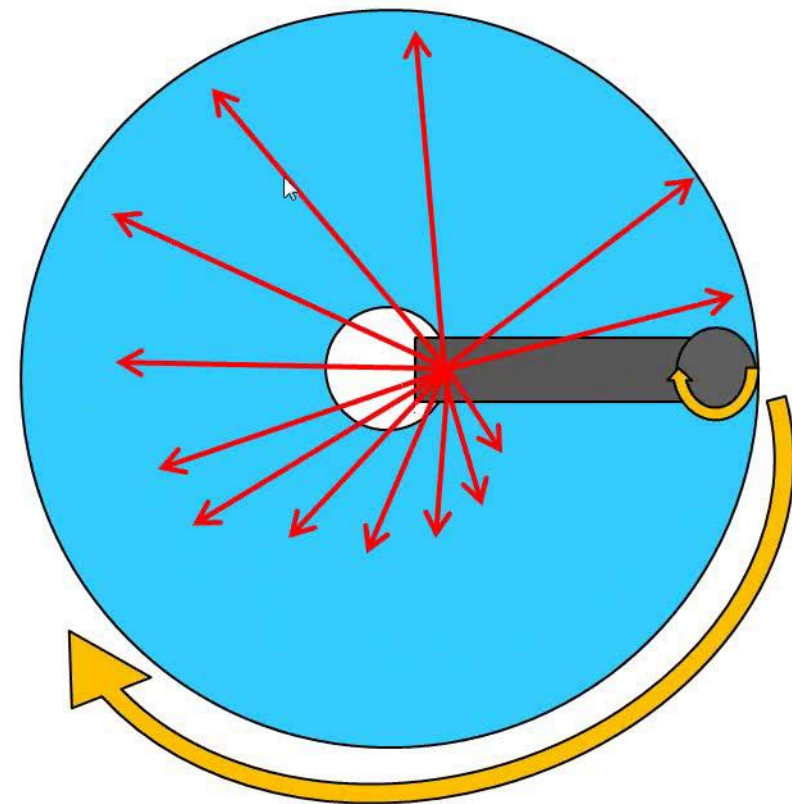
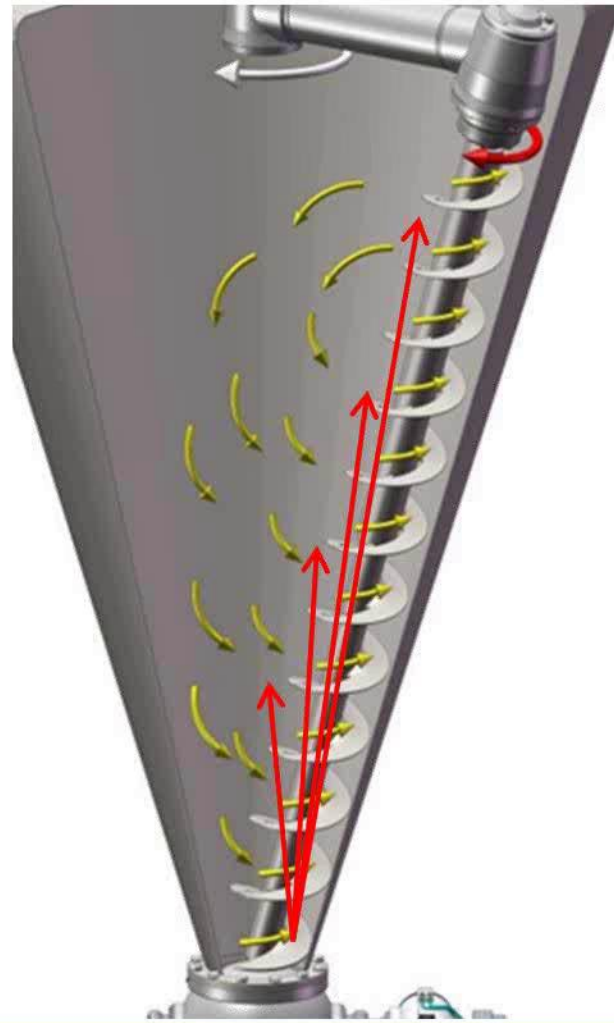
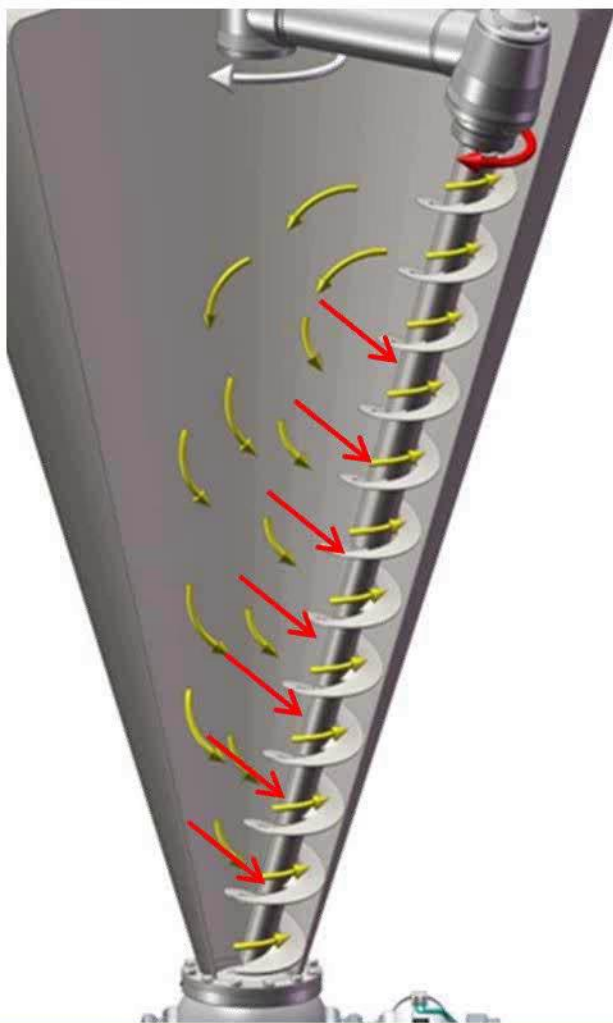


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Convective Transport by Auger



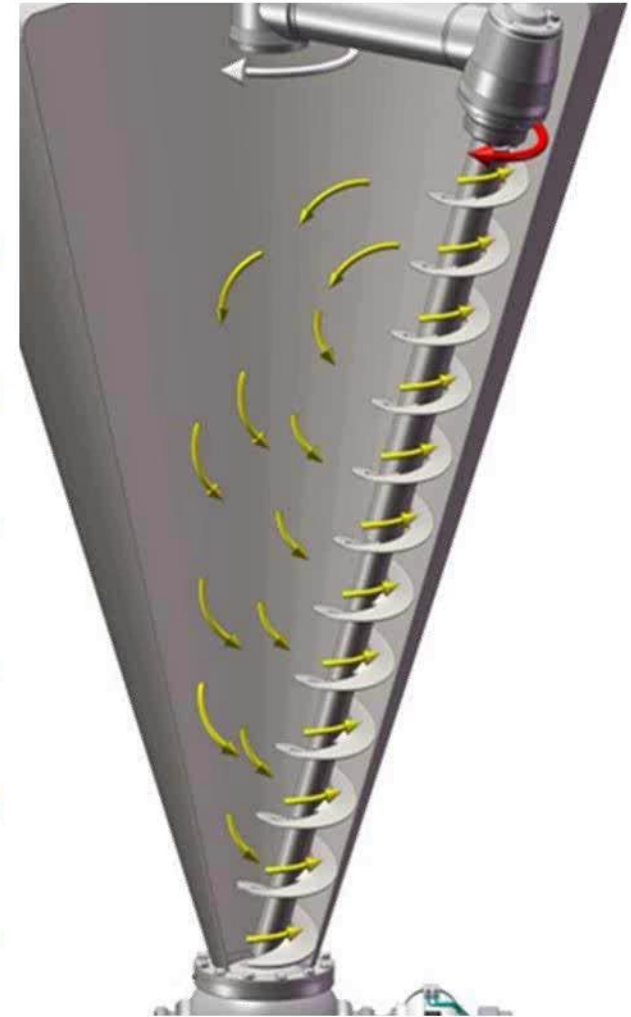
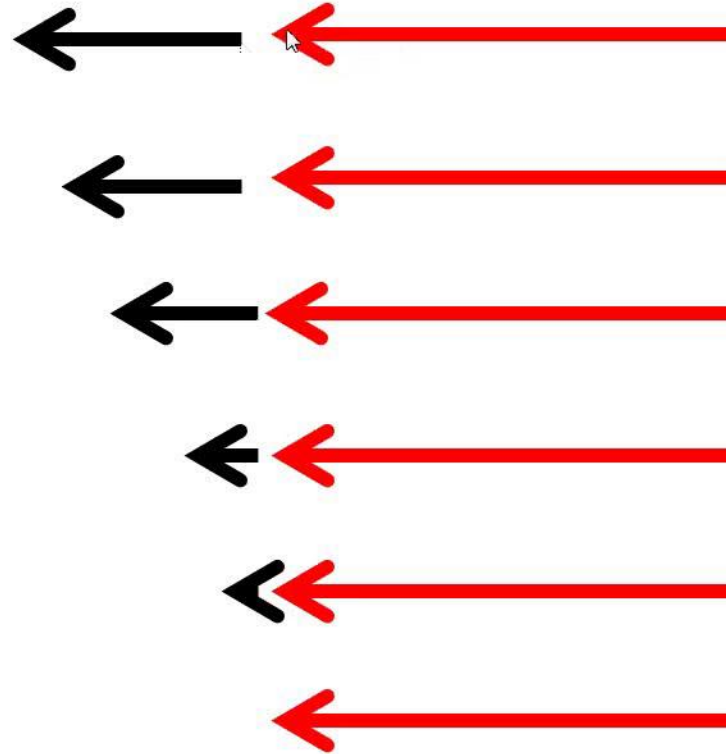
Controlled Randomization



HOSOKAWA MICRON POWDER SYSTEMS

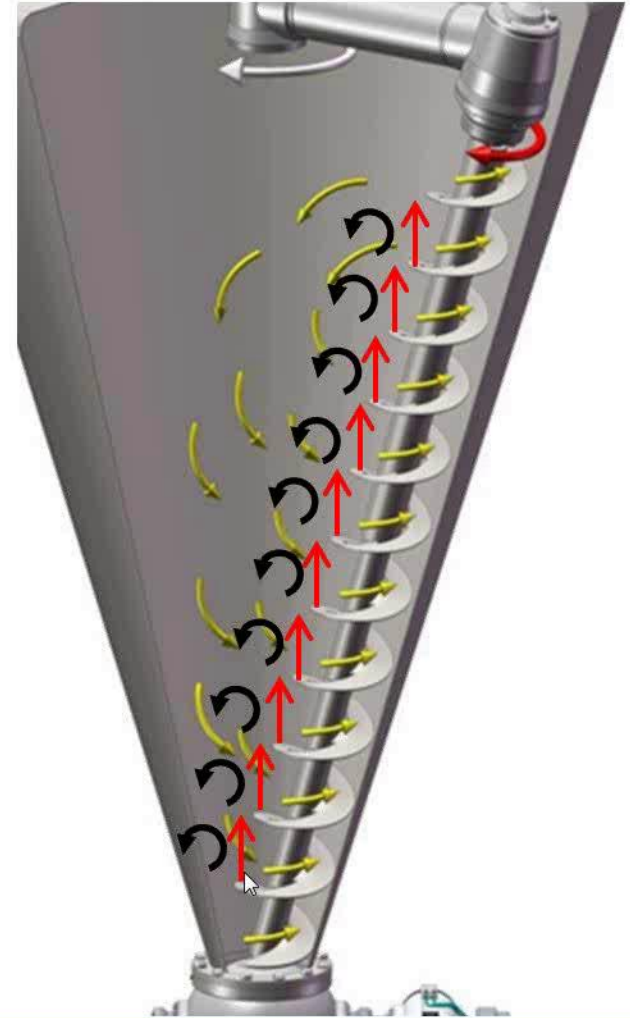
Teamwork = Auger + Orbital Arm

Tip speed
1.0-1.2 m/s
combined
speed

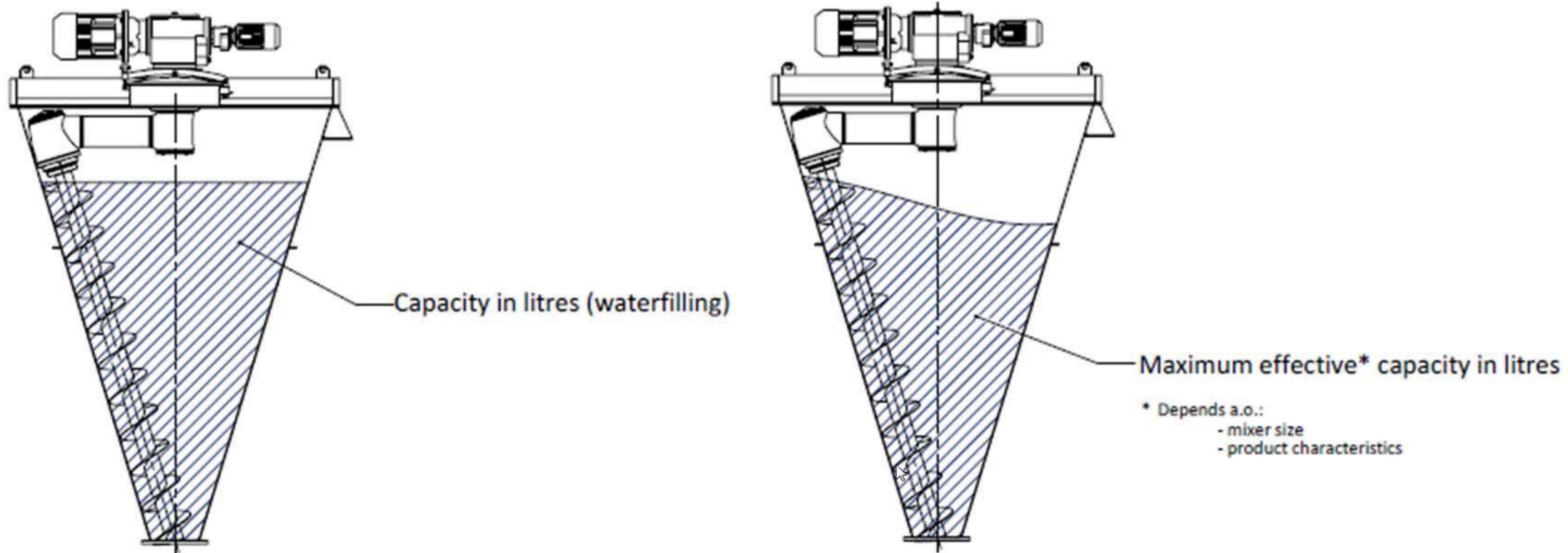


Teamwork = Auger + Orbital Arm

- Auger does more “work” than Orbital arm
- 10:1 ratio of motor sizes



Filling the Mixer



Mixer Capacity is 85-15% of water fill volume



HOSOKAWA MICRON POWDER SYSTEMS

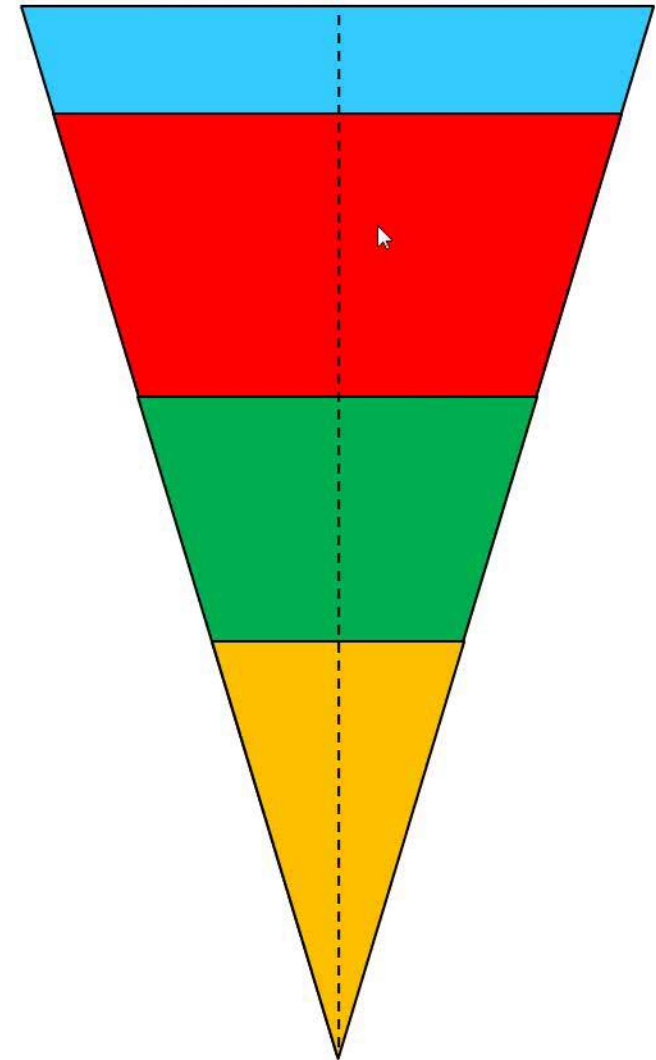
Video of Mixing



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Conical Shape Vessel

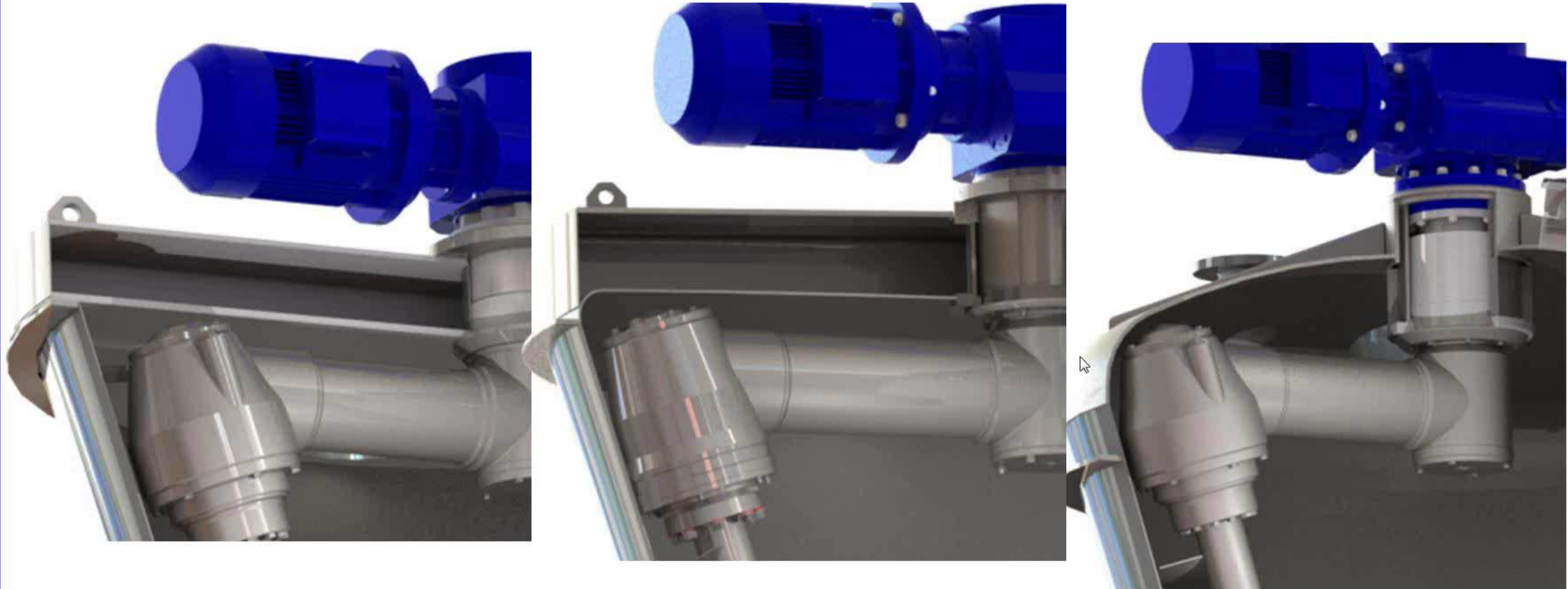
- $V = \pi * r^2 * h / 3$
 - With the bottom angle fixed $r \propto h$
 - $V \propto h^3$, $h \propto V^{1/3}$
 - Double height $\Rightarrow$ 8x more volume
 - In practice it is 8-10x more volume if you double the height



34° (17° from vertical)

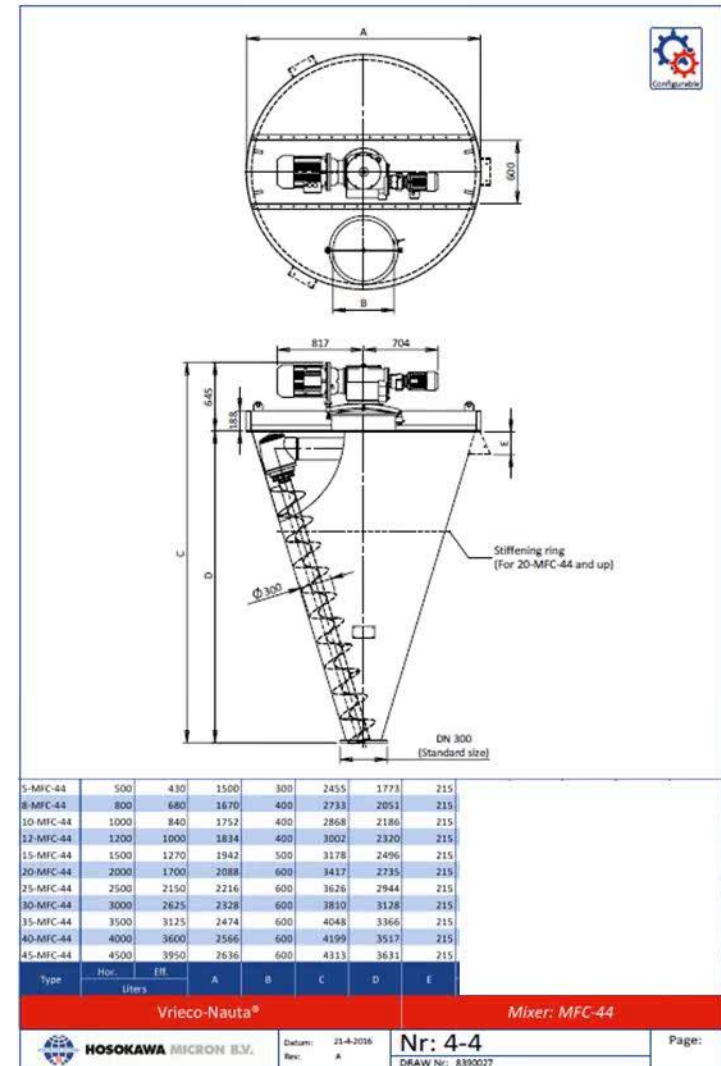


Top Cover



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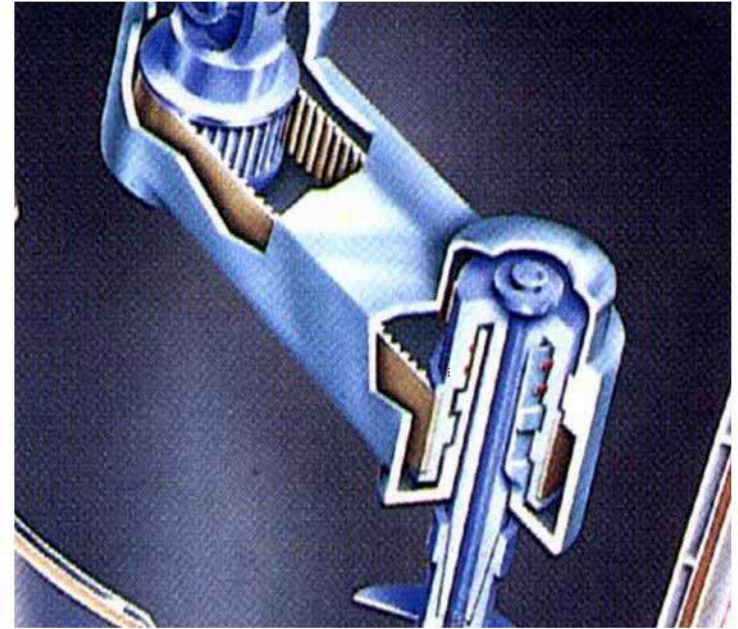
Drives



HOSOKAWA MICRON POWDER SYSTEMS

Orbital arm

Gear or belt drive
Shape determine type!



HOSOKAWA MICRON POWDER SYSTEMS

Orbital Arm Sealing Mechanisms

- Different types for different uses
- Two sets of seals
- Types:
 - Lip seal
 - Single Mech.
 - Double Mech.

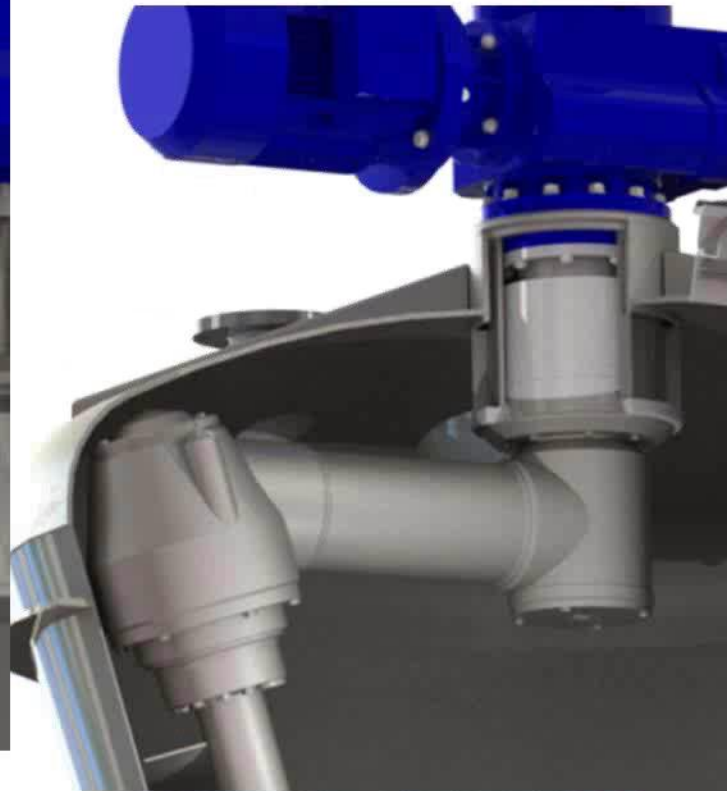
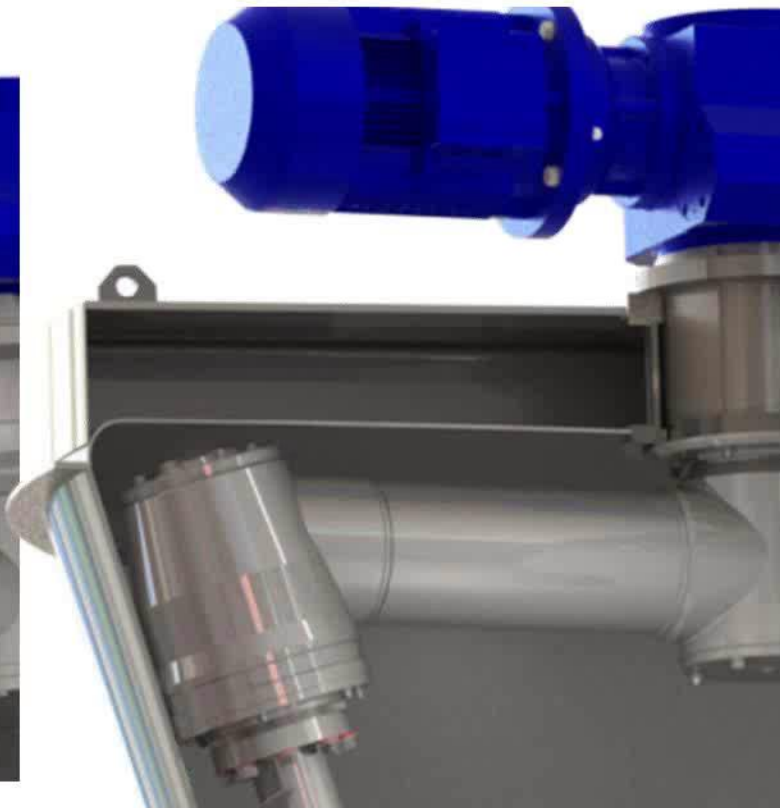


Mixing Screw (Auger)



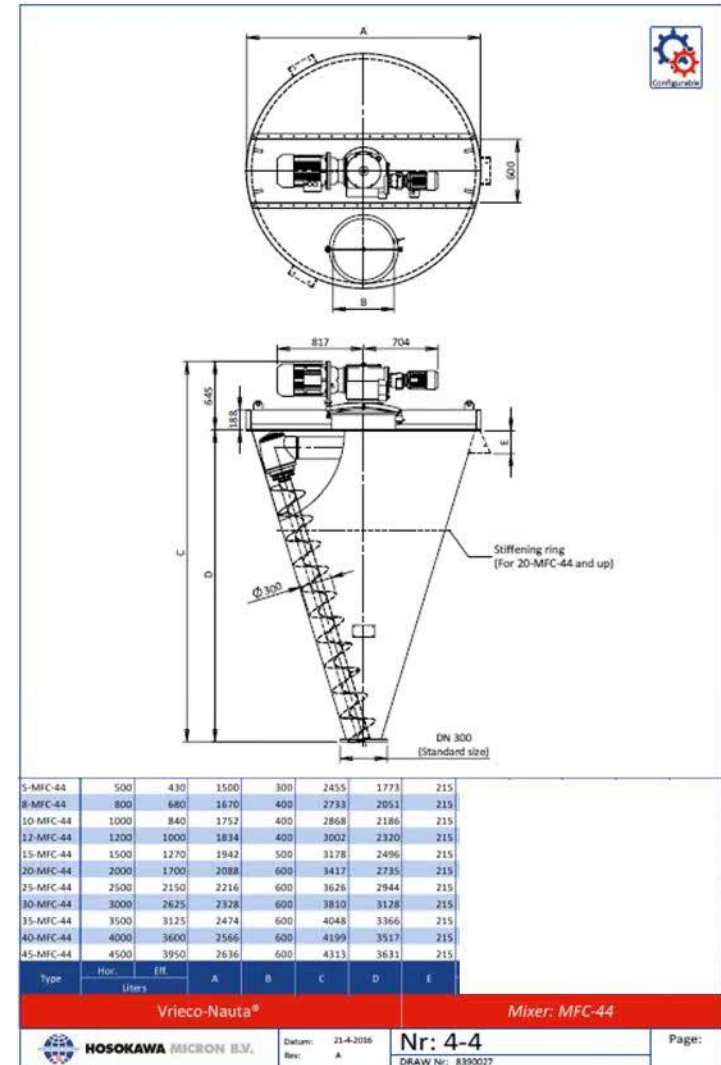
HOSOKAWA MICRON POWDER SYSTEMS

Top Cover



HOSOKAWA MICRON POWDER SYSTEMS

Drives



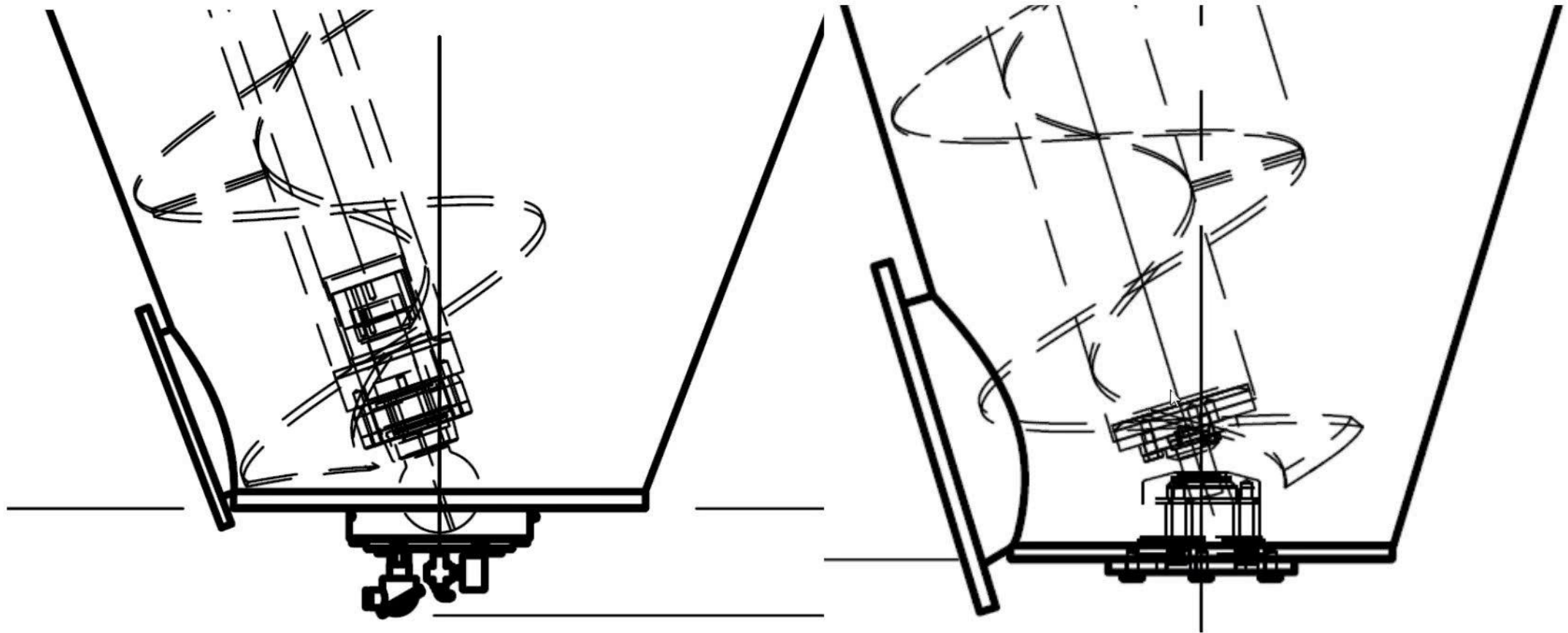
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Mixing Screw (Auger)



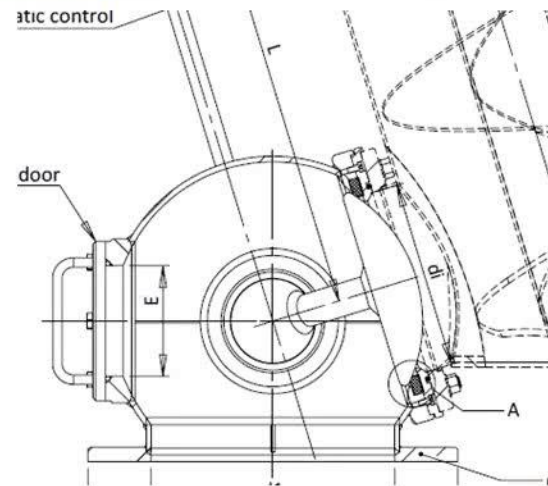
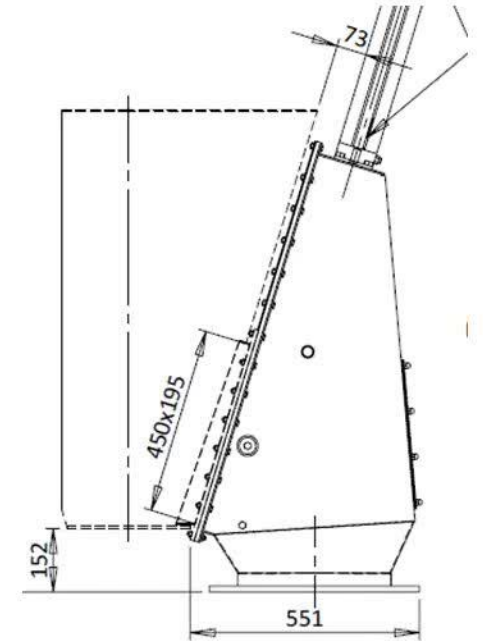
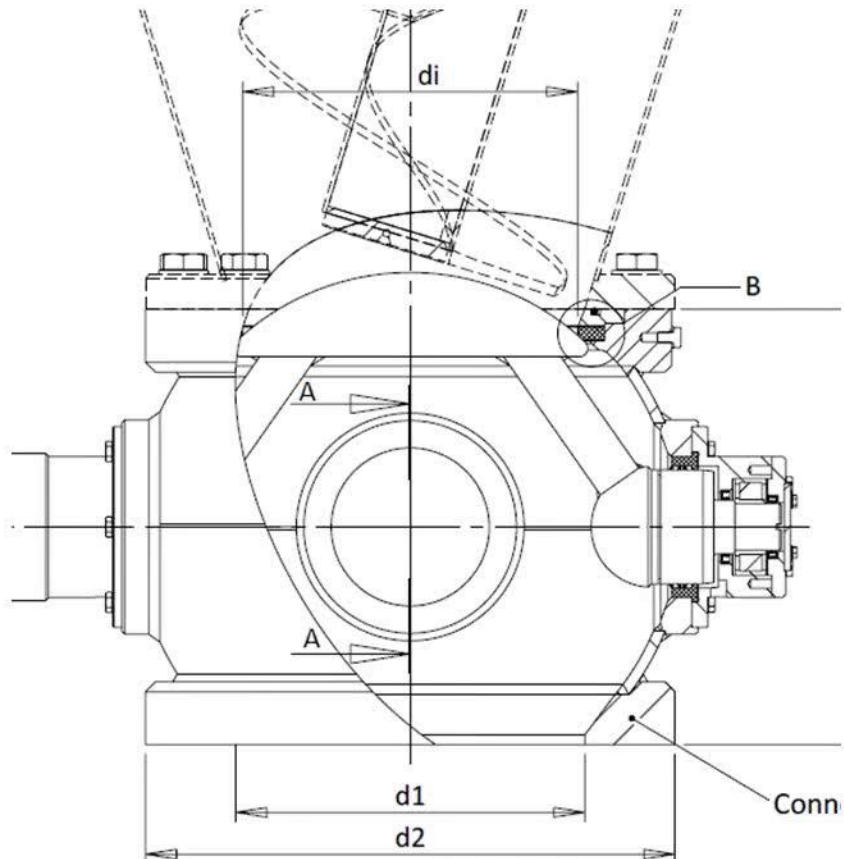
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Supported Mixing Screw



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Discharge Valves

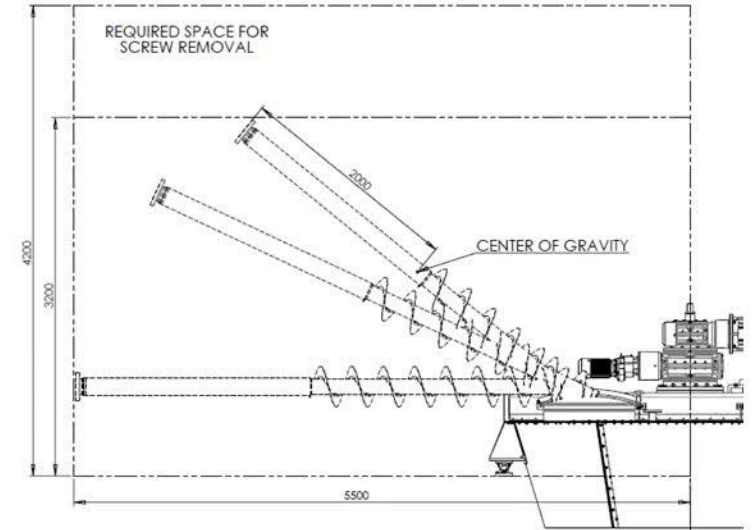
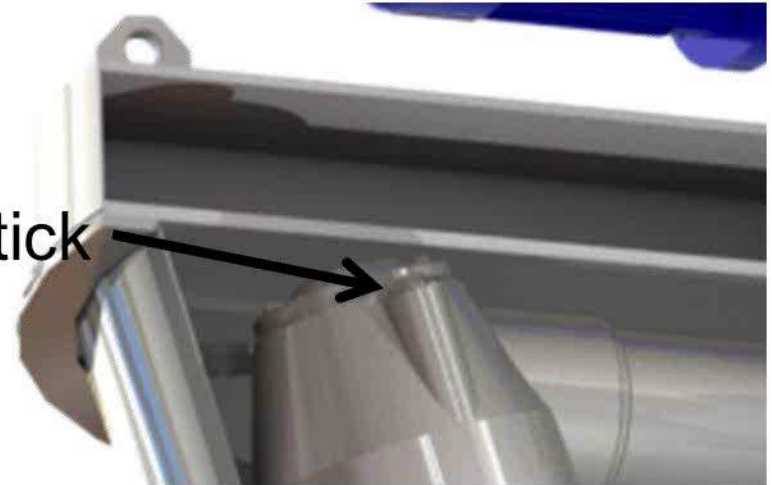


HOSOKAWA MICRON POWDER SYSTEMS

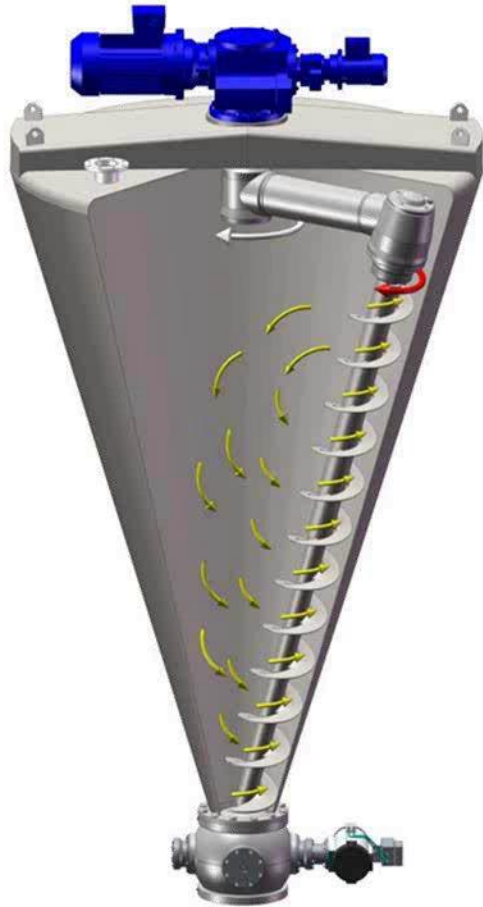
Maintenance

- Check Lubricants
- Change bearings/seals
 - Check height above unit!
 - Removal of top cover sections
 - Removal of entire bridge beam
 - Through inspection hatch
- Service personnel stationed in USA and Europe

Dipstick



Using the Nauta design for drying

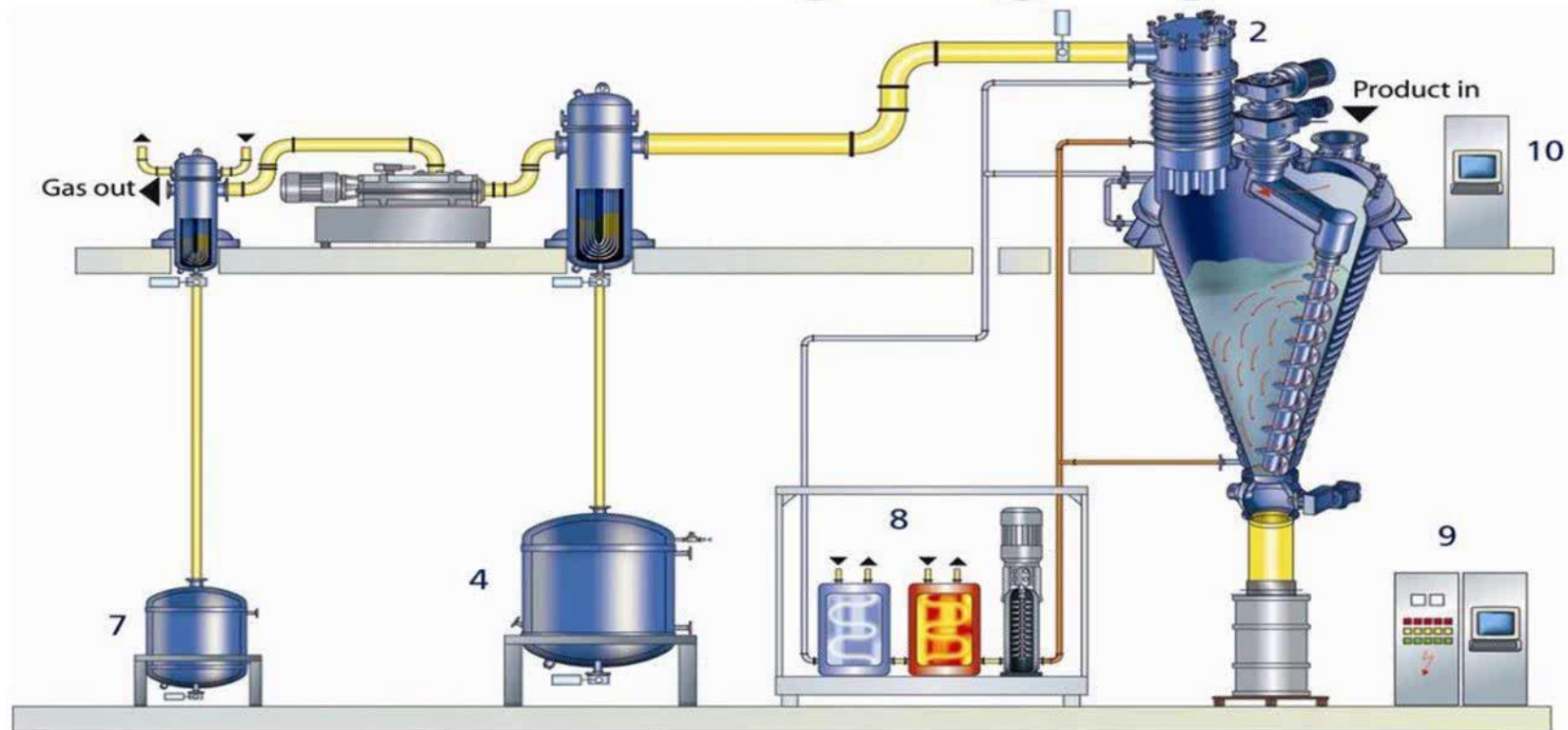


Changes
to



HOSOKAWA MICRON POWDER SYSTEMS

Vacuum drying system



1. Vacuum dryer
2. Vacuum filter
3. Condenser
4. Receptacle

5. Vacuum pump
6. Optional condenser
7. Optional receptacle
8. Heating cooling skid

9. Control cabinet
10. Operating panel

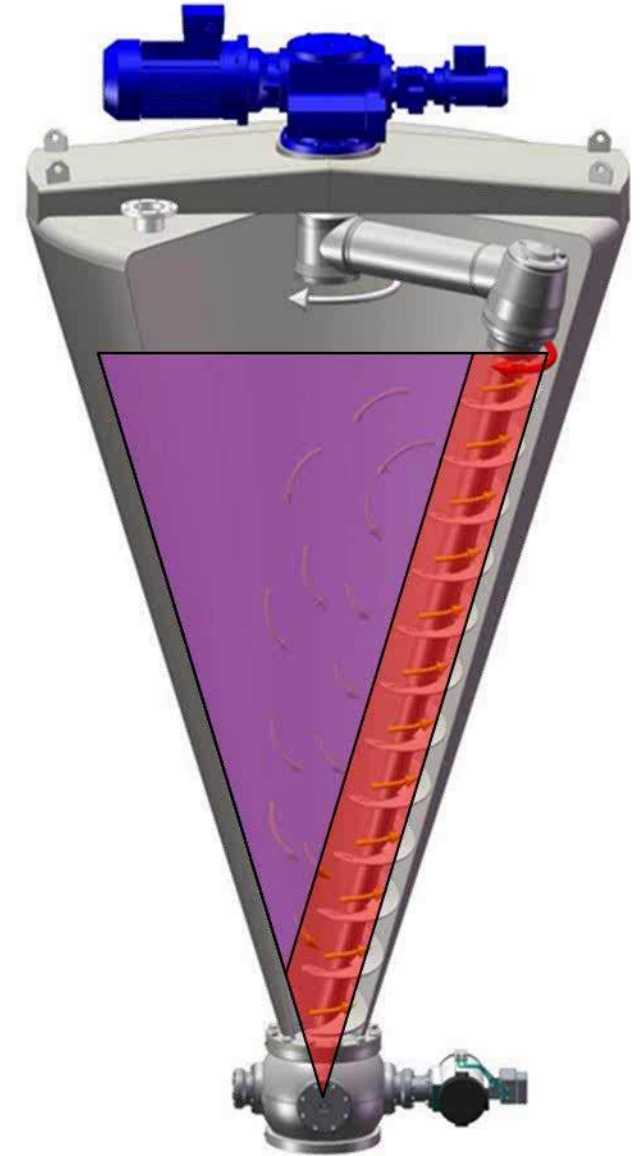


HOSOKAWA MICRON POWDER SYSTEMS

Metal Powders

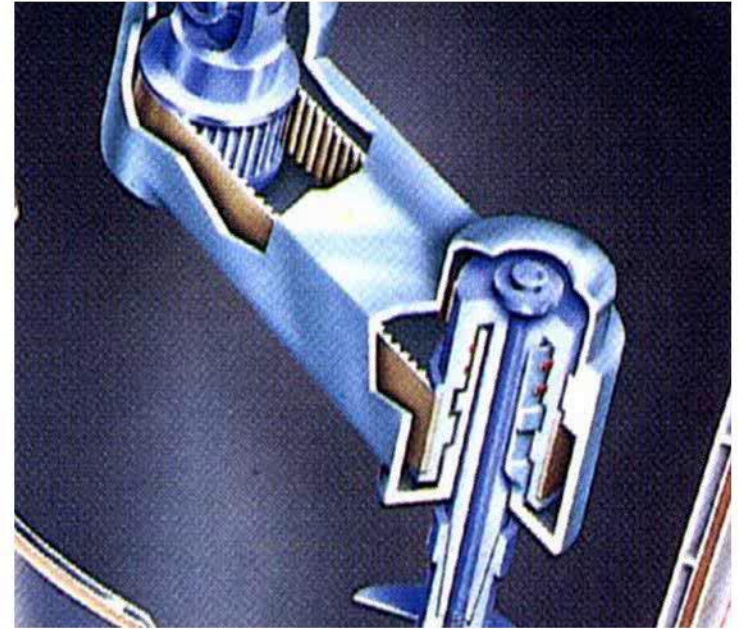


- UO₂ and Fe base powder
- Bulk Density over 2.5 g/mL



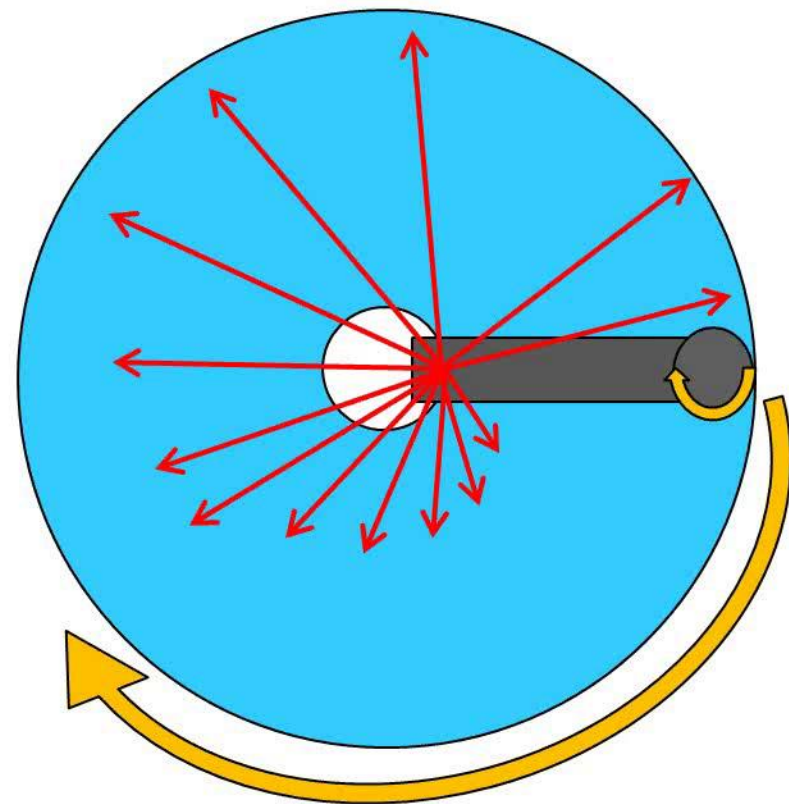
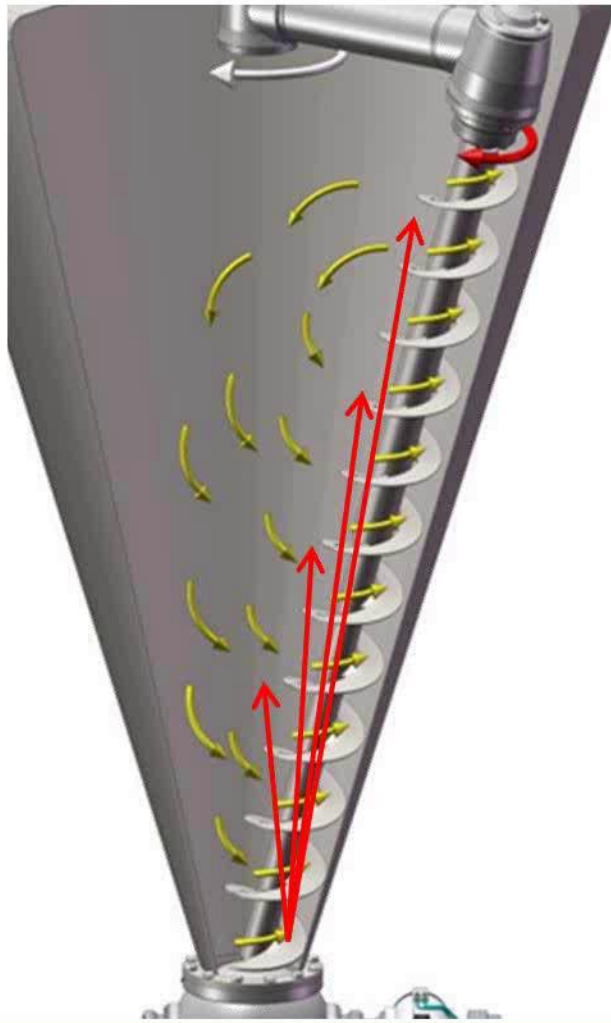
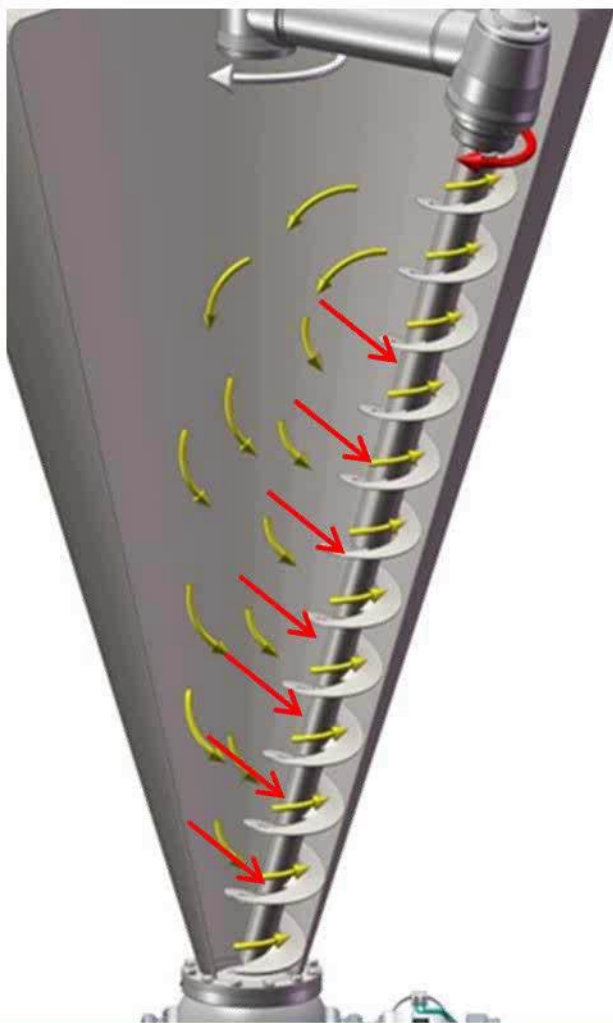
Orbital arm

Gear or belt drive
Shape determine type!



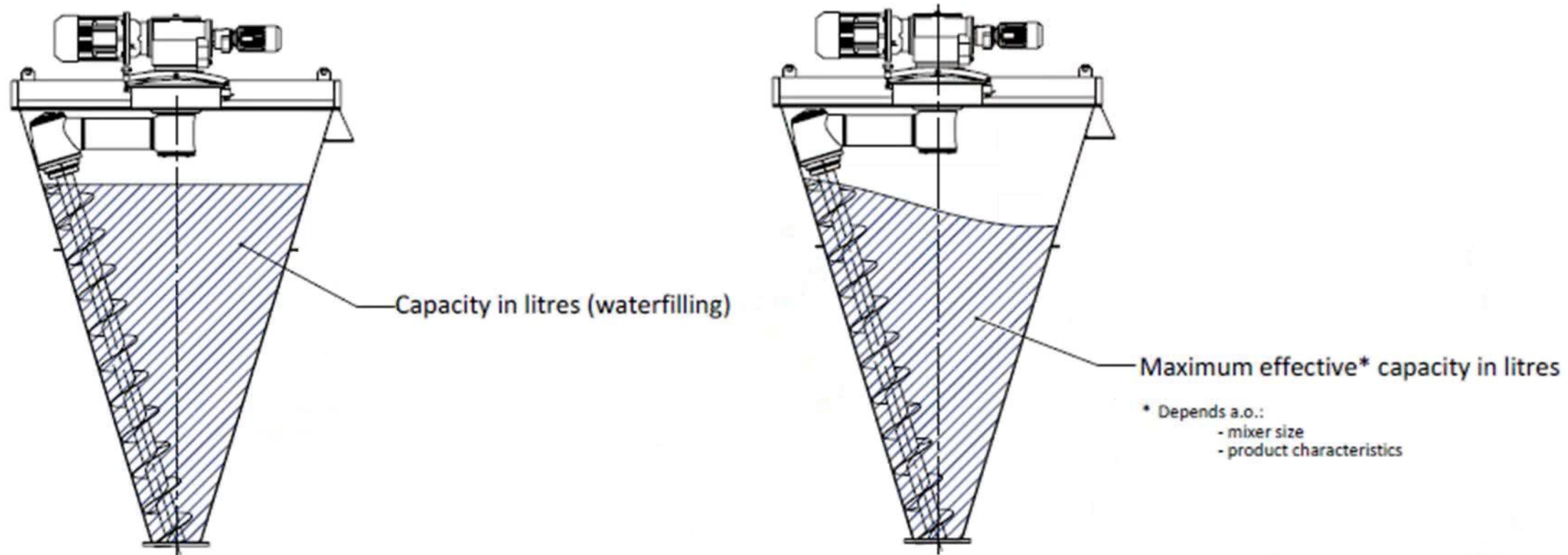
HOSOKAWA MICRON POWDER SYSTEMS

Controlled Randomization



HOSOKAWA MICRON POWDER SYSTEMS

Filling the Mixer



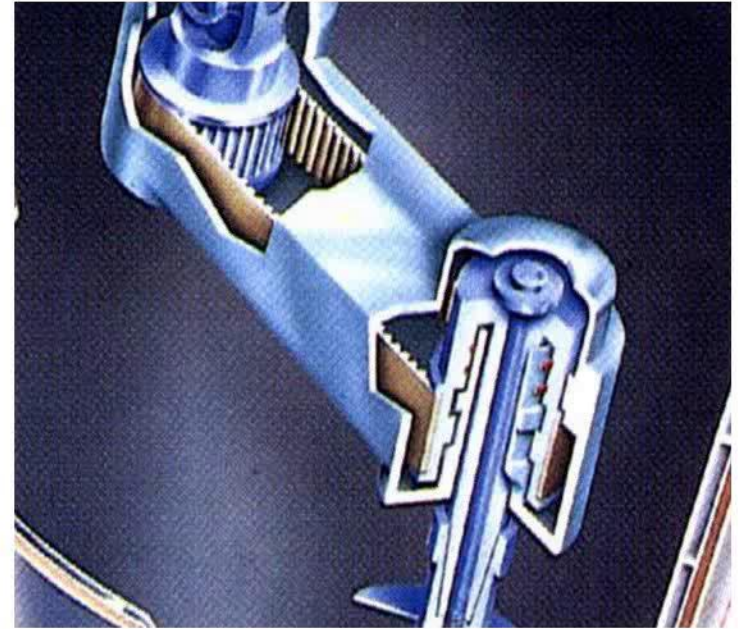
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HOSOKAWA MICRON POWDER SYSTEMS

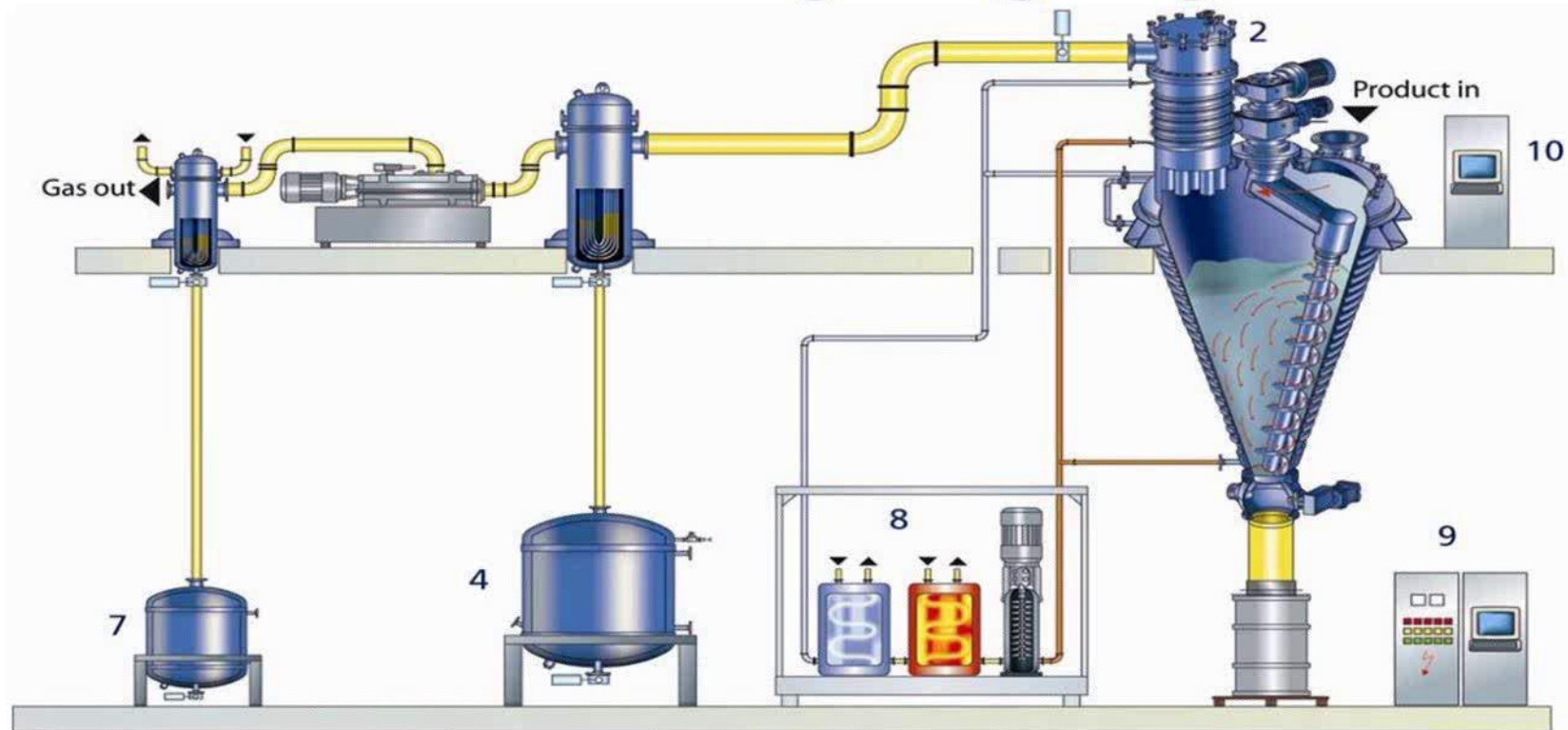
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HOSOKAWA MICRON POWDER SYSTEMS

Vacuum drying system



1. Vacuum dryer
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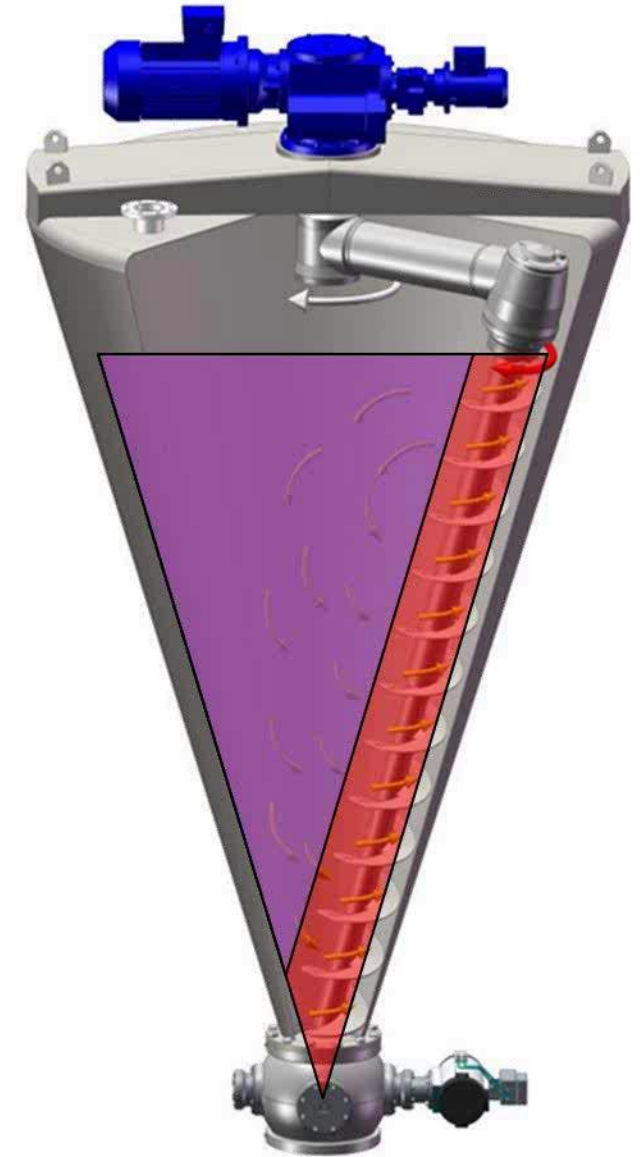


HOSOKAWA MICRON POWDER SYSTEMS

Metal Powders

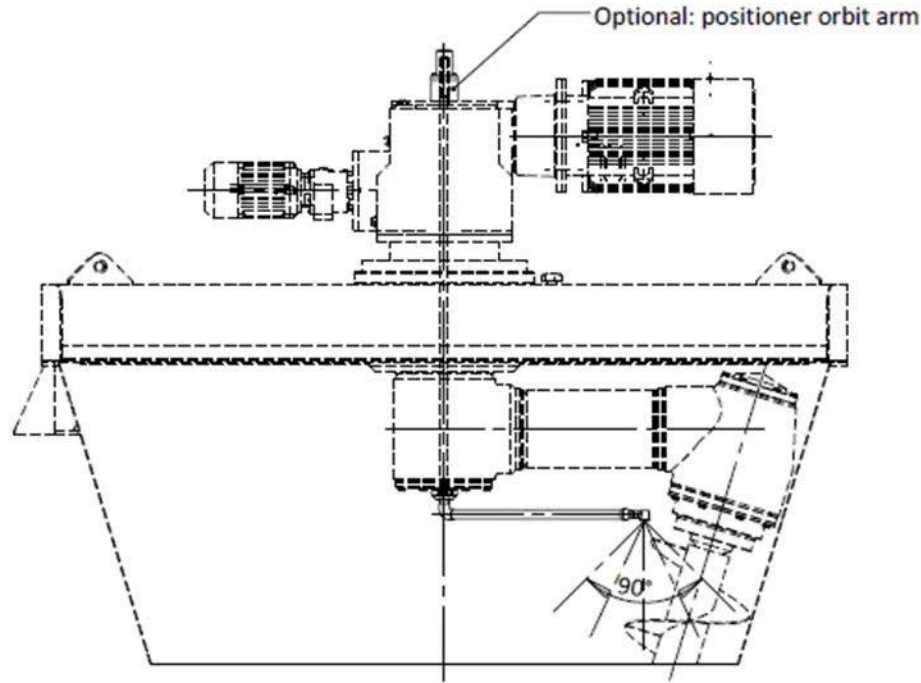


- UO₂ and Fe base powder
- Bulk Density over 2.5 g/mL

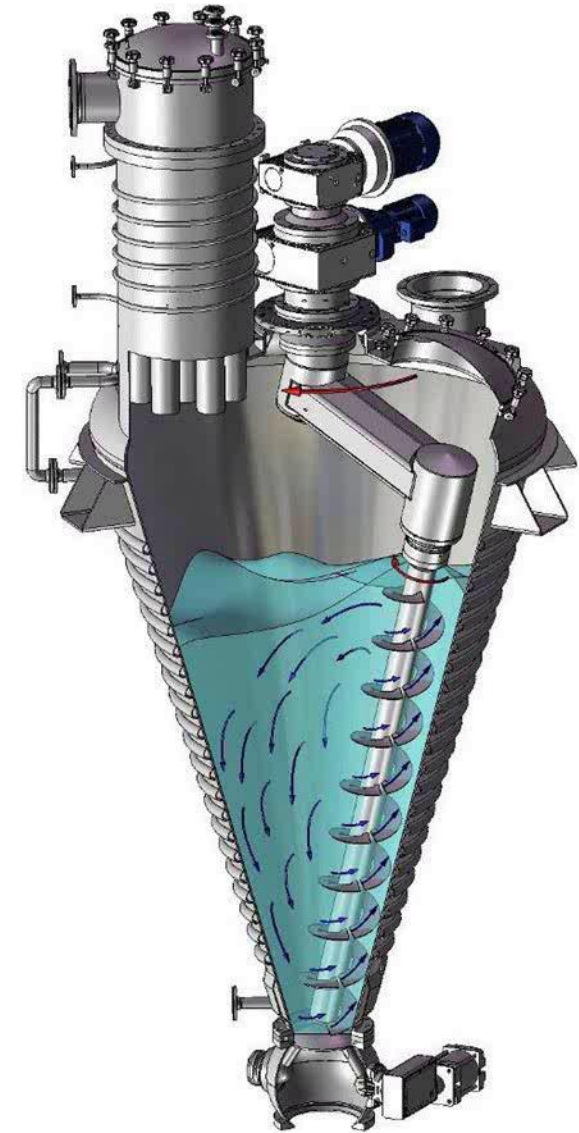


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Friable Catalyst

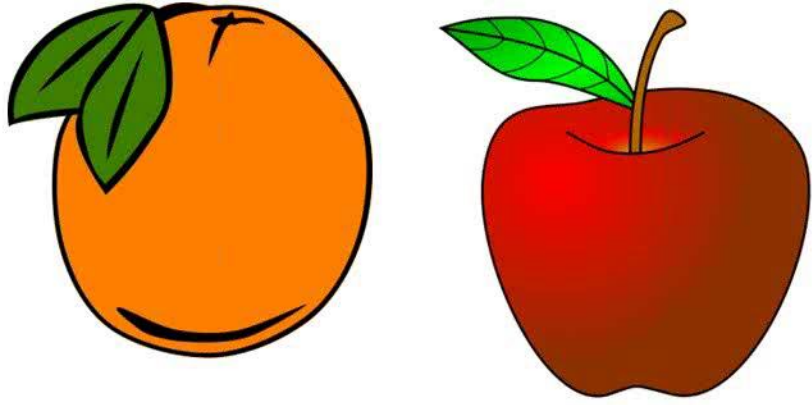


Coating and Drying in a
single vessel

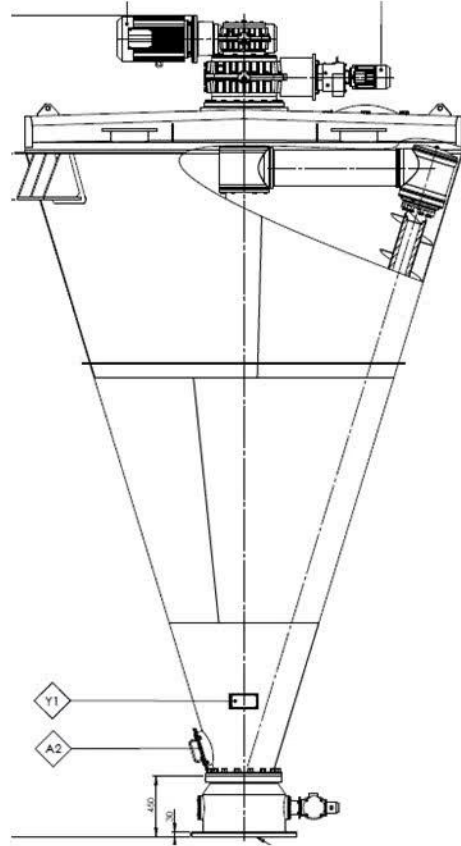


HOSOKAWA MICRON POWDER SYSTEMS

Pectin Blending



- Gentle Blending
- Minimal heat generation even for large batches





9-1 Fixed WIP_CIP dual-+-8390096_1.PDF - Adobe Acrobat Pro

File Edit View Window Help

Create

1 / 1

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Tools

Comment

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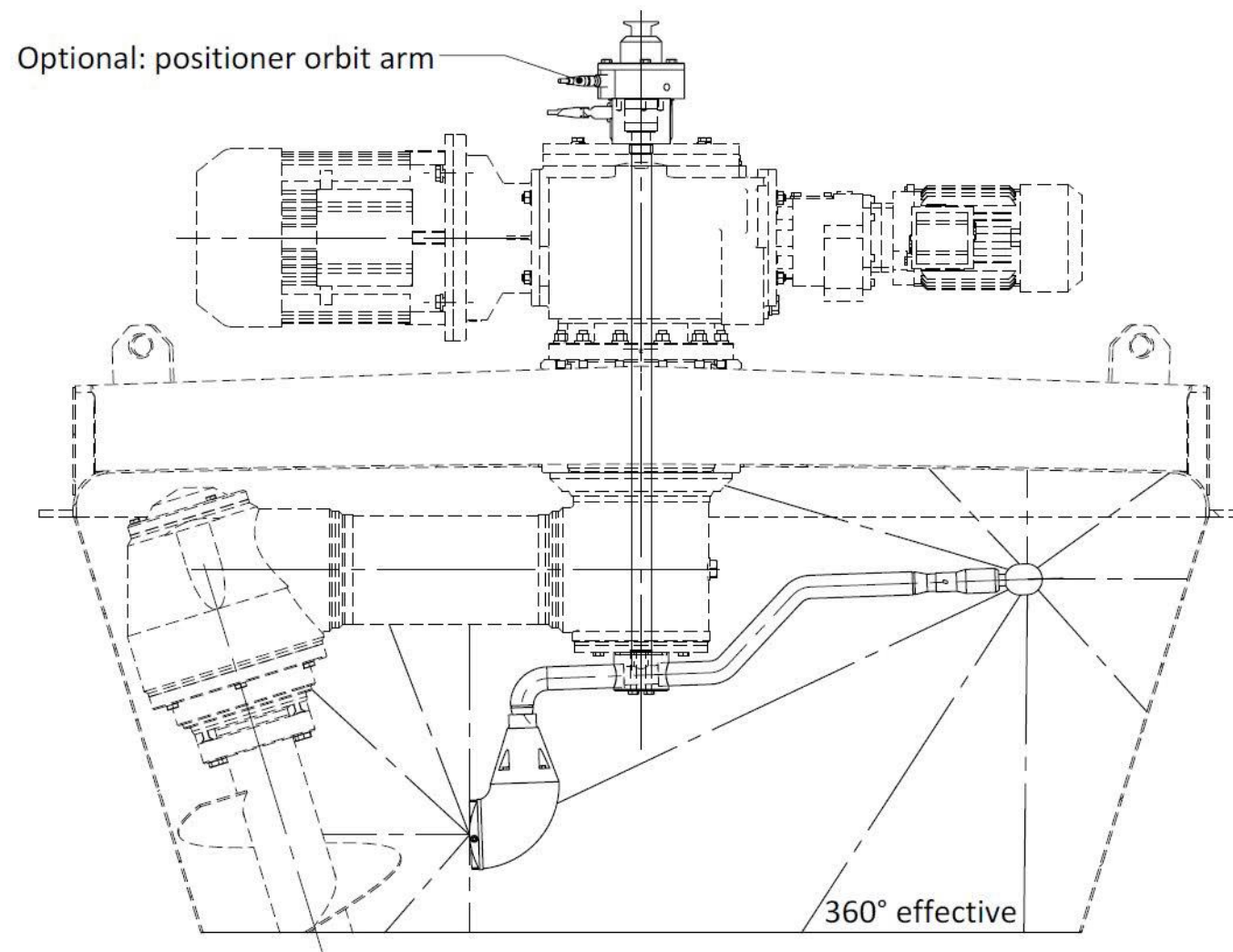
Optional: positioner orbit arm

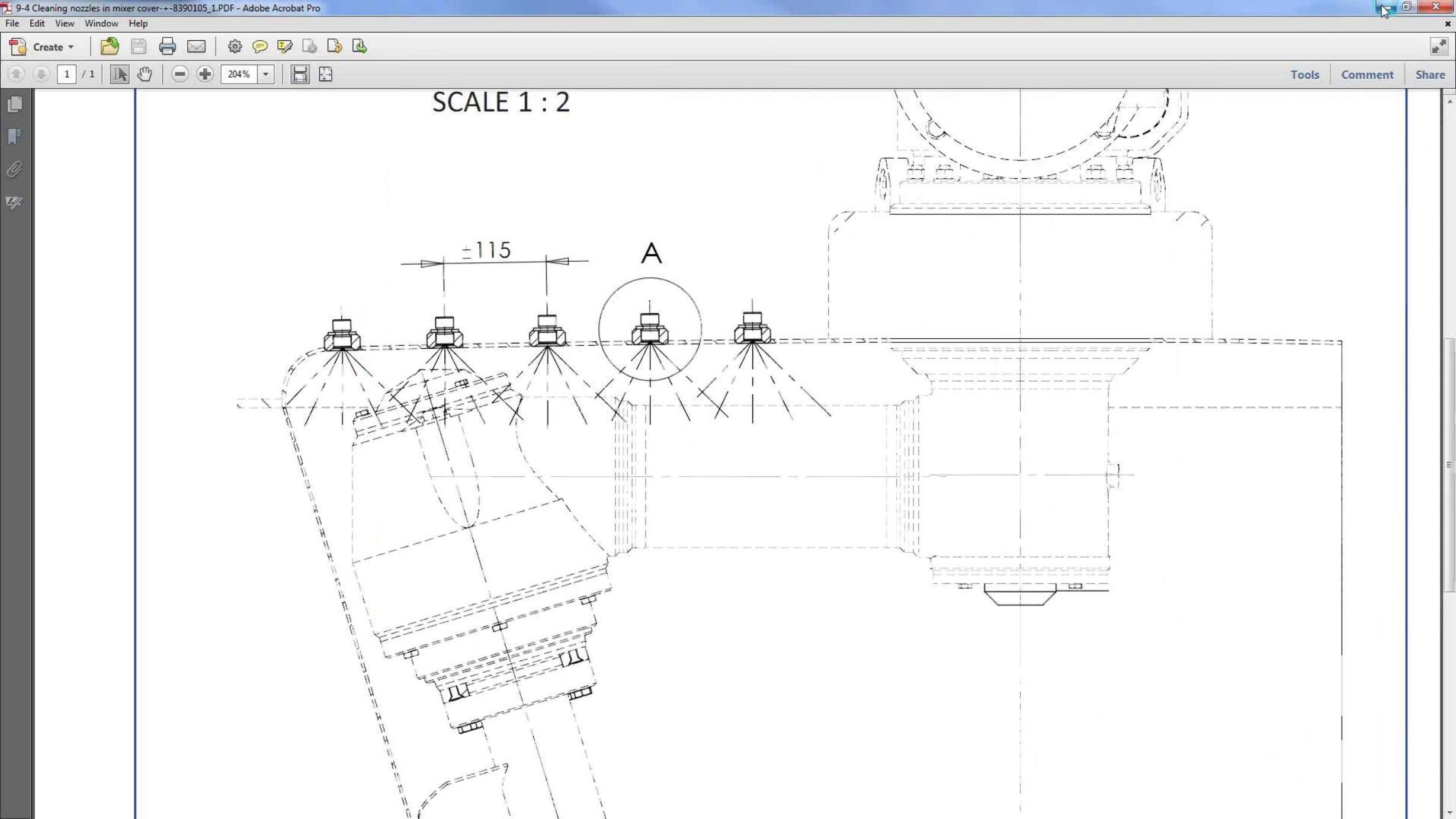
360° effective

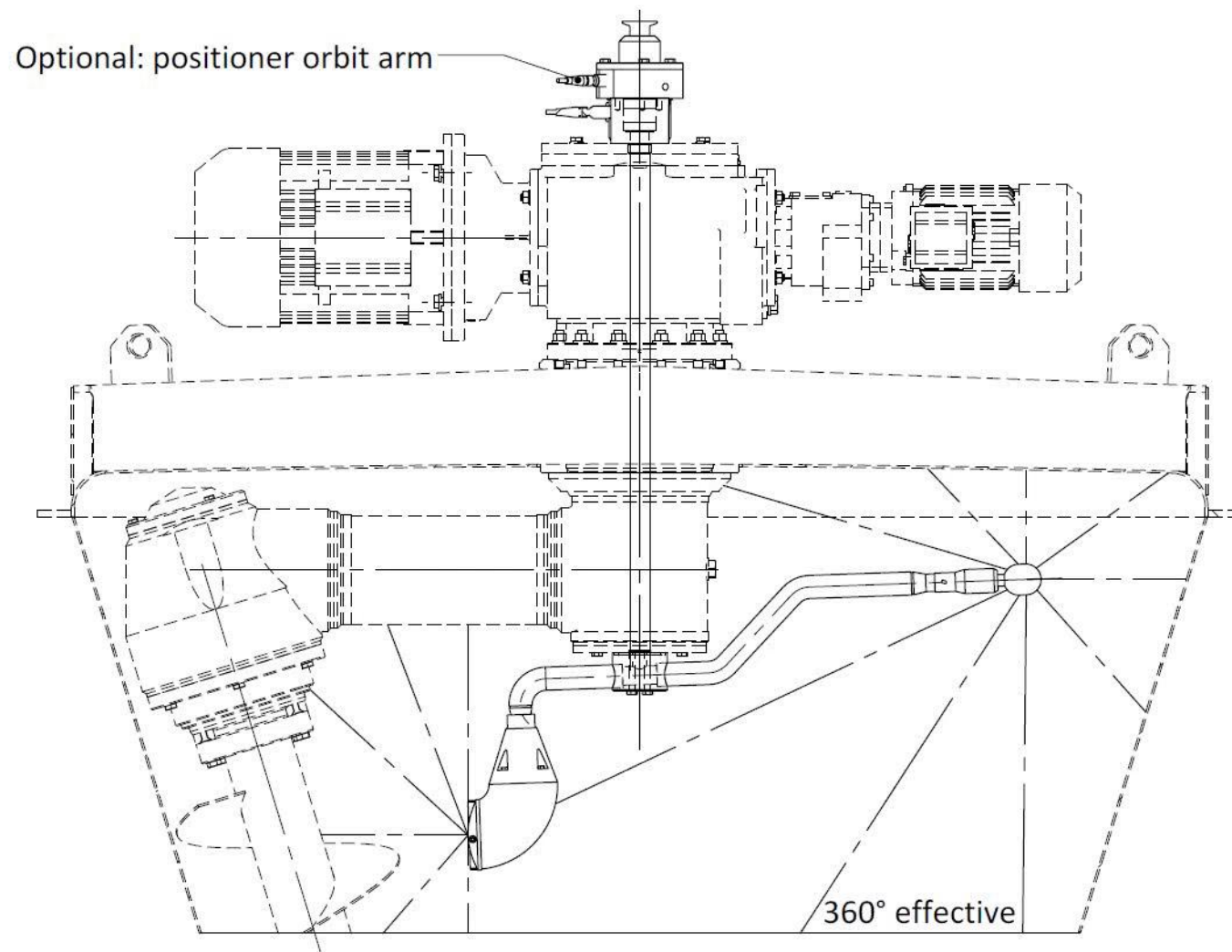
Flow rates are indicative, depending on the type of mixer and volume

Flow (at 6 bar)	[l/min]	± 140	Flow (at 2 bar)	[l/min]	± 100
Working Pressure	[bar]	3 - 8	Working Pressure	[bar]	1 - 3

Configuration B

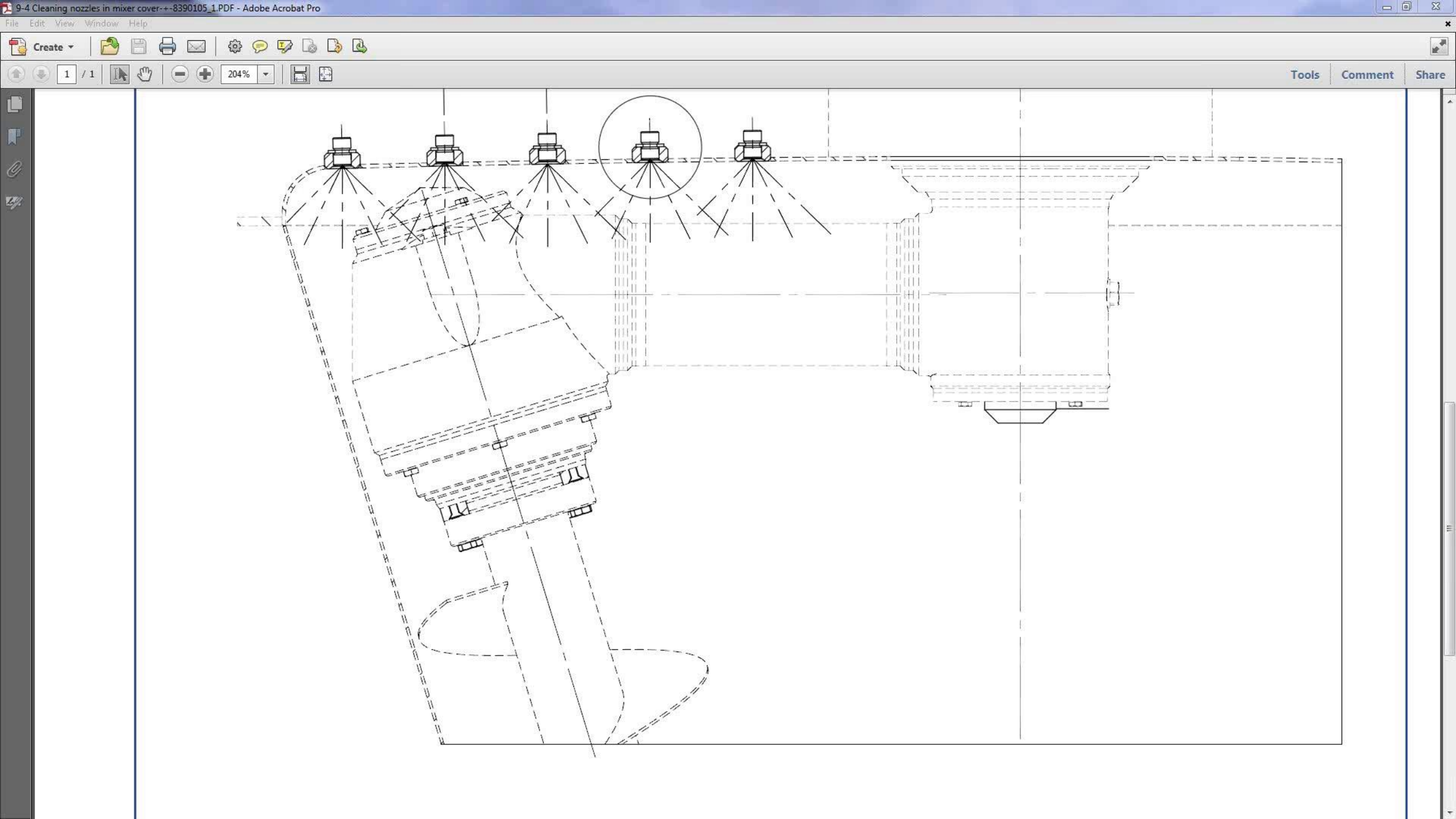




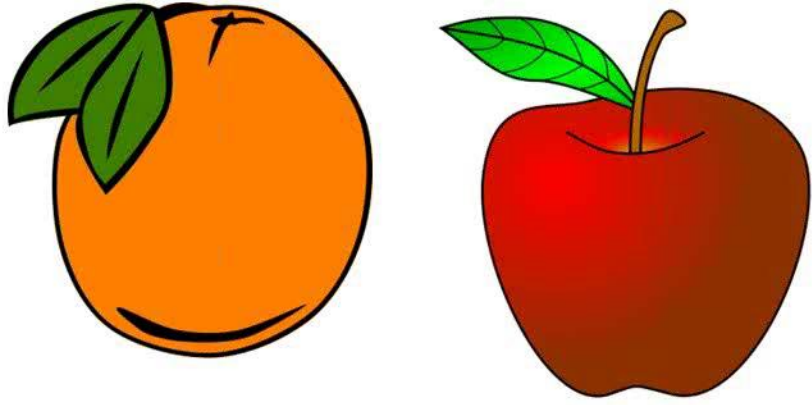


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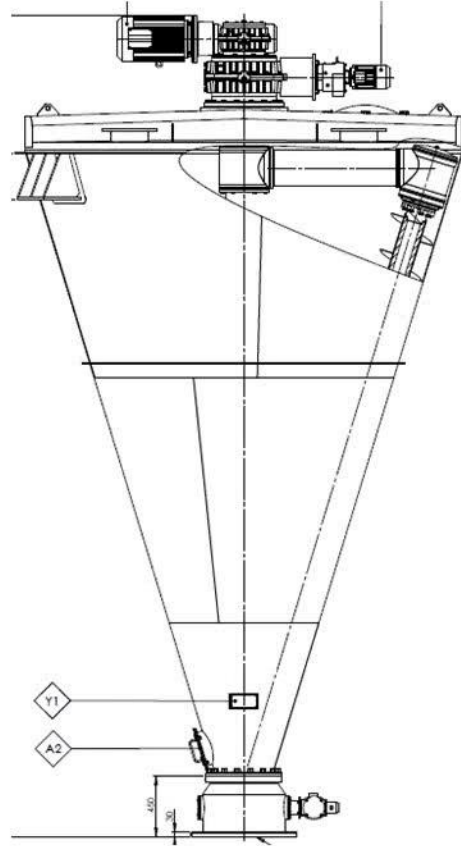




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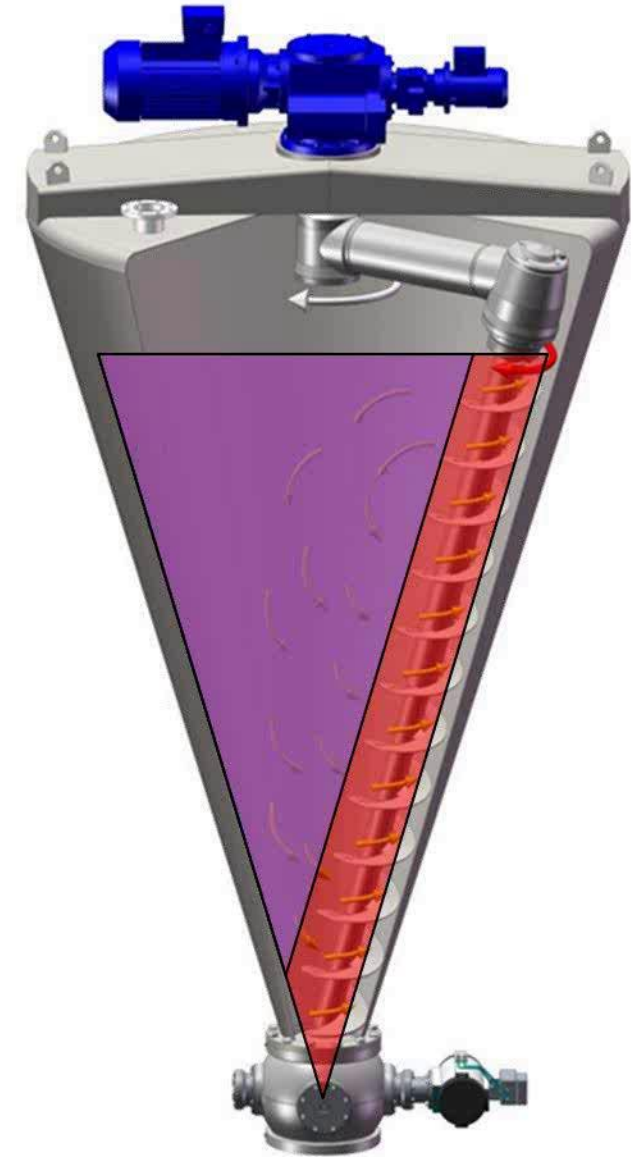
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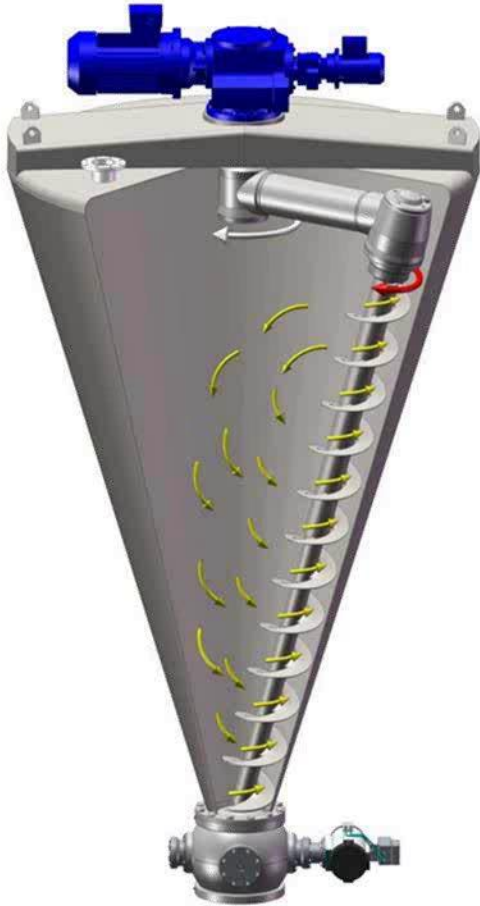


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- Bulk Density over 2.5 g/mL



HOSOKAWA MICRON POWDER SYSTEMS

Using the Nauta design for drying



Changes
to



HOSOKAWA MICRON POWDER SYSTEMS

Thank you for your attention!

Hosokawa Micron Powder Systems
10 Chatham Road
Summit, NJ 07901 USA

Chris Paulsworth

Phone: 908-277-9272

cpaulsworth@hmps.hosokawa.com



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